

RIVER WATER QUALITY IN HONG KONG IN 2018



Environmental Protection Department
The Government of the Hong Kong Special Administrative Region

Mission

To conduct a comprehensive and scientific monitoring programme that helps safeguard the health of Hong Kong's rivers and streams and achieve the Water Quality Objectives.



River Indus

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1. Introduction

Rivers and streams in Hong Kong are generally short with small flows, many of which have their upstream sections located within the catchment areas of reservoirs where potable water is gathered. Other than for drinking water supply, the primary beneficial uses of the rivers and streams in Hong Kong include maintenance of aquatic life, as general amenity, flood prevention and storm water drainage. The main channel of Shing Mun River is the only watercourse used for secondary contact recreation activities.

To provide data for water pollution control and management purposes, the Environmental Protection Department (EPD) has implemented a comprehensive river water quality monitoring programme since 1986. It mainly covers major rivers flowing through the urban areas, with one or more representative monitoring stations set up at the upstream and downstream sections of the main channels and tributaries. A number of small rivers and streams in the New Territories are also monitored.

The monitoring programme serves the following purposes:

- evaluate the pollution status of river water;
- monitor long-term changes in water quality;
- provide a scientific basis for planning water pollution control strategies;
- assess compliance with the Water Quality Objectives (WQOs); and
- compile Water Quality Index (WQI).

This report summarises the water quality of rivers and streams covered by EPD's river monitoring programme in 2018. Annual reports can be downloaded from the EPD's website:

<https://www.epd.gov.hk/epd/english/environmentinhk/water/hkwqrc/waterquality/river-2.html>



Flow Measurement at Tai Wai Nullah

2. An Overview of Hong Kong's Rivers and Streams in 2018

In 2018, the EPD's river monitoring programme covered 30 watercourses and a total of 82 monitoring stations (Appendix A), as compared with some 14 watercourses and 47 stations when the programme started in 1986. The monitoring programme involved monthly field measurements and water sample collection for laboratory analyses. Over 50 physico-chemical and biological parameters, including organic matters, nutrients, metals and *E. coli*, were analysed (Appendix B).

2.1. Compliance Rate of Water Quality Objectives (WQOs)

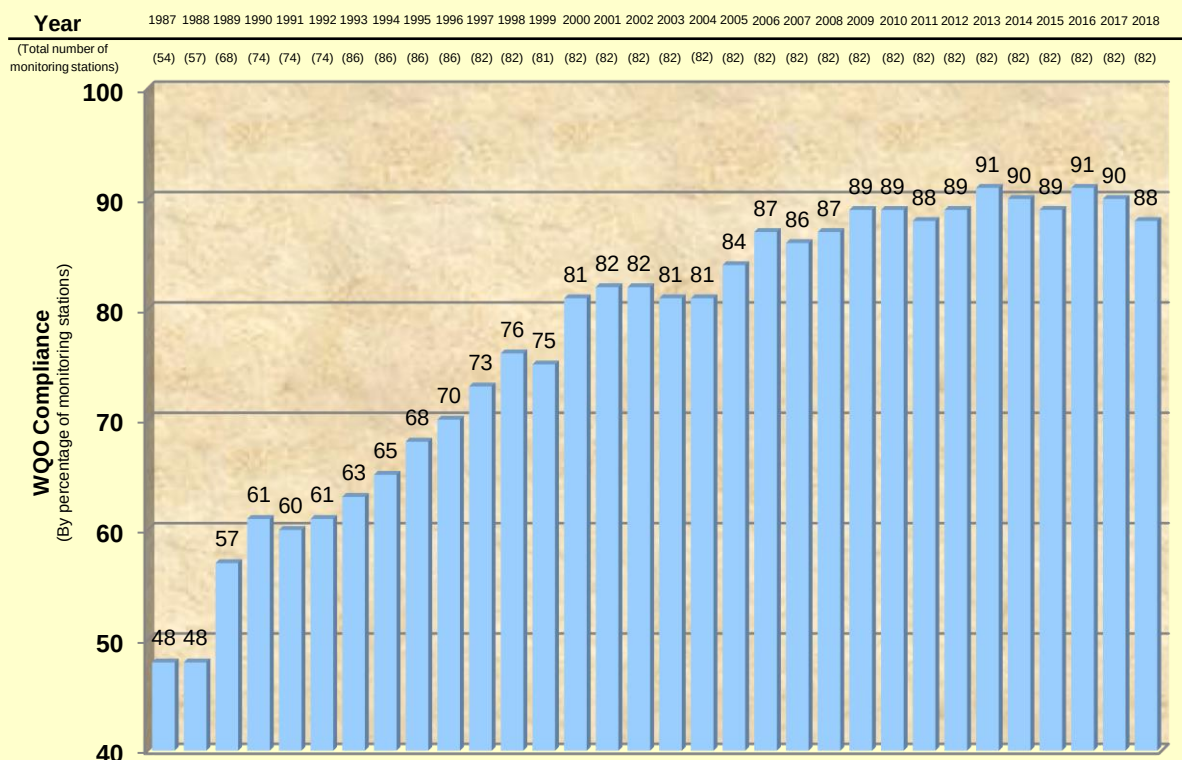
To assess compliance with the Water Quality Objectives (WQOs) for individual monitoring stations, 5 representative parameters are used, namely pH, suspended solids, dissolved oxygen, 5-day biochemical oxygen demand (BOD₅) and chemical oxygen demand (COD) (Appendix C). This report presents the annual average compliance rates for individual watercourses as well as overall compliance rates for all rivers and streams in Hong Kong (Appendix F).

For rivers that have more than one monitoring station, the WQO compliance was expressed as an average of the annual compliance rates of all their individual monitoring stations where full compliance would be indicated by 100%. Similarly, the overall compliance rates for the entire Hong Kong were the averages of that of all the monitoring stations in the territory. The river water quality in Hong Kong in 2018 continued to be good with an overall compliance rate at 88%, and within the range of fluctuations over the past ten years (87% – 91%) (Figure 1). Figure 2 shows the WQO compliance in the river waters for key river water quality parameters over the last three decades, 1987–2018.

The high compliance rates in recent years were the results of the implementation of pollution control legislation, including the Water Pollution Control Ordinance and the Livestock Waste Control Scheme introduced under the Waste Disposal Ordinance, as well as the extension of the sewer network to more villages under the Sewerage Master Plans (SMPs).

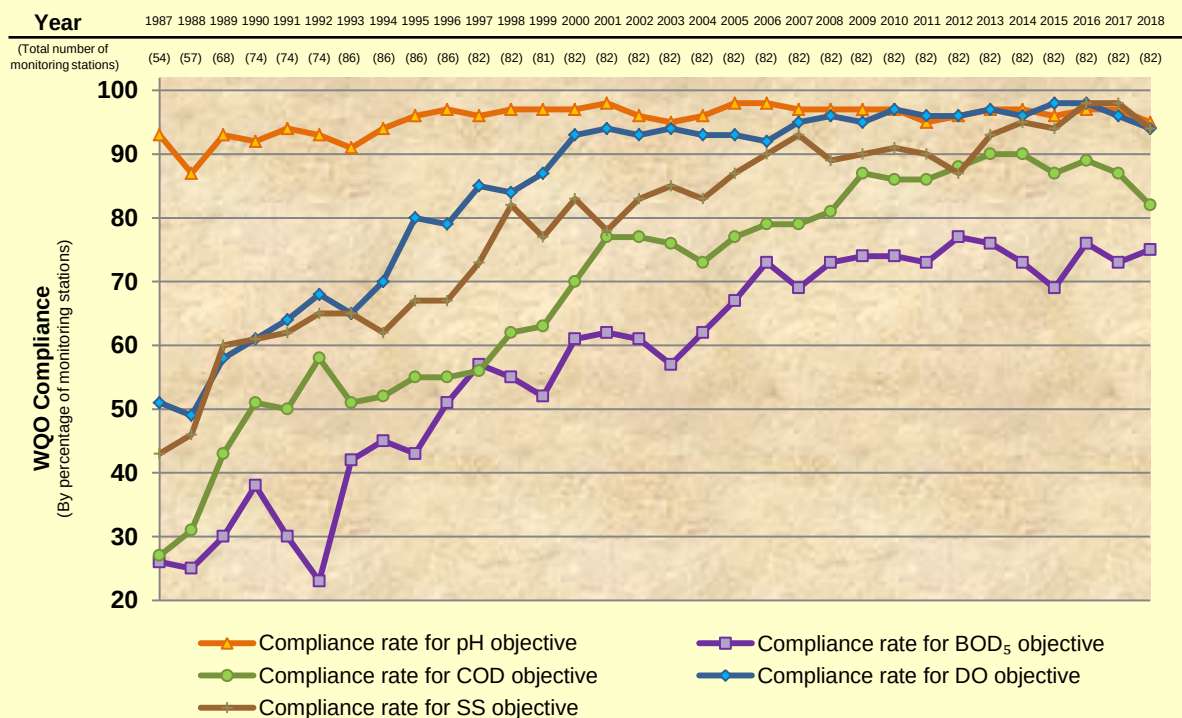
In summary, the state of water quality of Hong Kong's watercourses remained largely steady in 2018 and followed a long-term improvement trend as a result of the gradual reduction of pollution loading to the rivers and streams.

Figure 1. Overall WQO compliance in the river waters of Hong Kong, 1987-2018



Figures are rounded to the nearest integer

Figure 2. WQO compliance for five key parameters in the river waters of Hong Kong, 1987-2018



2.2. Water Quality Index (WQI) Gradings

The Water Quality Index (WQI) is relevant to conserving the primary beneficial use of rivers and streams for supporting aquatic life. It grades the general ecological health of rivers and streams into five categories of “Excellent”, “Good”, “Fair”, “Bad” and “Very Bad” by assessing three key water quality parameters: dissolved oxygen, 5-day biochemical oxygen demand (BOD₅) and ammonia-nitrogen content¹.

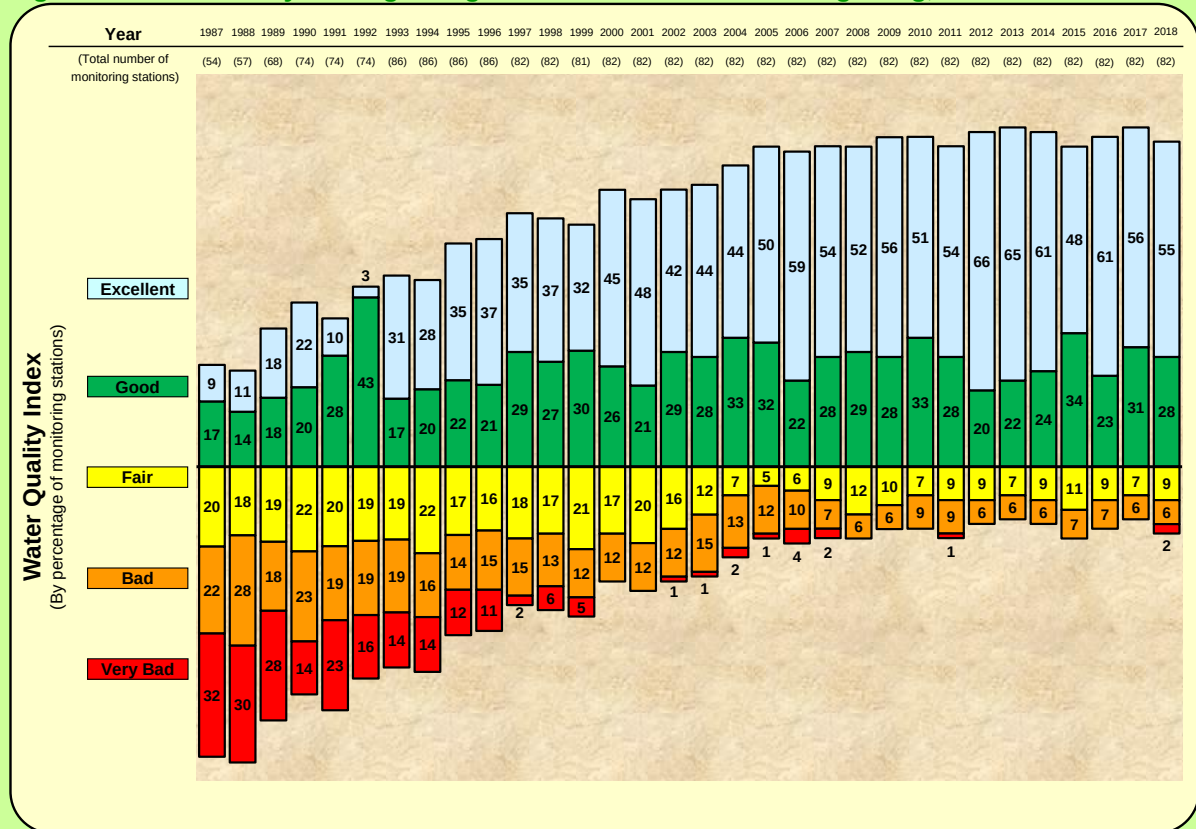
In 2018, 83% of the river monitoring stations were graded “Good” or “Excellent”, as compared with only 26% in 1987 (Figure 3), suggesting that the river water quality has greatly improved and the pollution loadings in these watercourses have been significantly reduced over the past three decades. Majority of the monitoring stations in Lantau Island, the Eastern New Territories, the Southwestern New Territories and Kowloon were graded in these two categories. For comparison, only 8% of the monitoring stations were graded “Bad” or “Very Bad” in 2018, while 54% “Bad” or “Very Bad” were recorded in 1987. Most of the “Bad” or “Very Bad” stations are located along watercourses in the Northwestern New Territories (Figures 4 & 5). Minor changes of gradings were observed in certain stations in 2018 which are within the normal range of natural fluctuations over the past ten years (Table 1).

Table 1. Stations with WQI gradings in 2018 changed as compared to 2017

River monitoring station	2017 WQI grading	2018 WQI grading
Shing Mun River (TR19)	Good	Excellent
Ho Chung River (PR1)	Good	Excellent
Kam Tin River (KT1)	Bad	Fair
Tseng Lan Shue Stream (JR3)	Fair	Bad
River Beas (RB1)	Excellent	Good
River Beas (RB3)	Good	Fair
River Ganges (GR1, GR2)	Good	Fair
Yuen Long Creek (YL1, YL2)	Fair	Bad
Yuen Long Creek (YL3, YL4)	Bad	Very Bad
Tuen Mun River (TN4, TN5)	Excellent	Good

¹ Please refer to Appendix G for more details on the calculation and assessment of Water Quality Index.

Figure 3. Water Quality Index gradings for the river waters of Hong Kong, 1987-2018

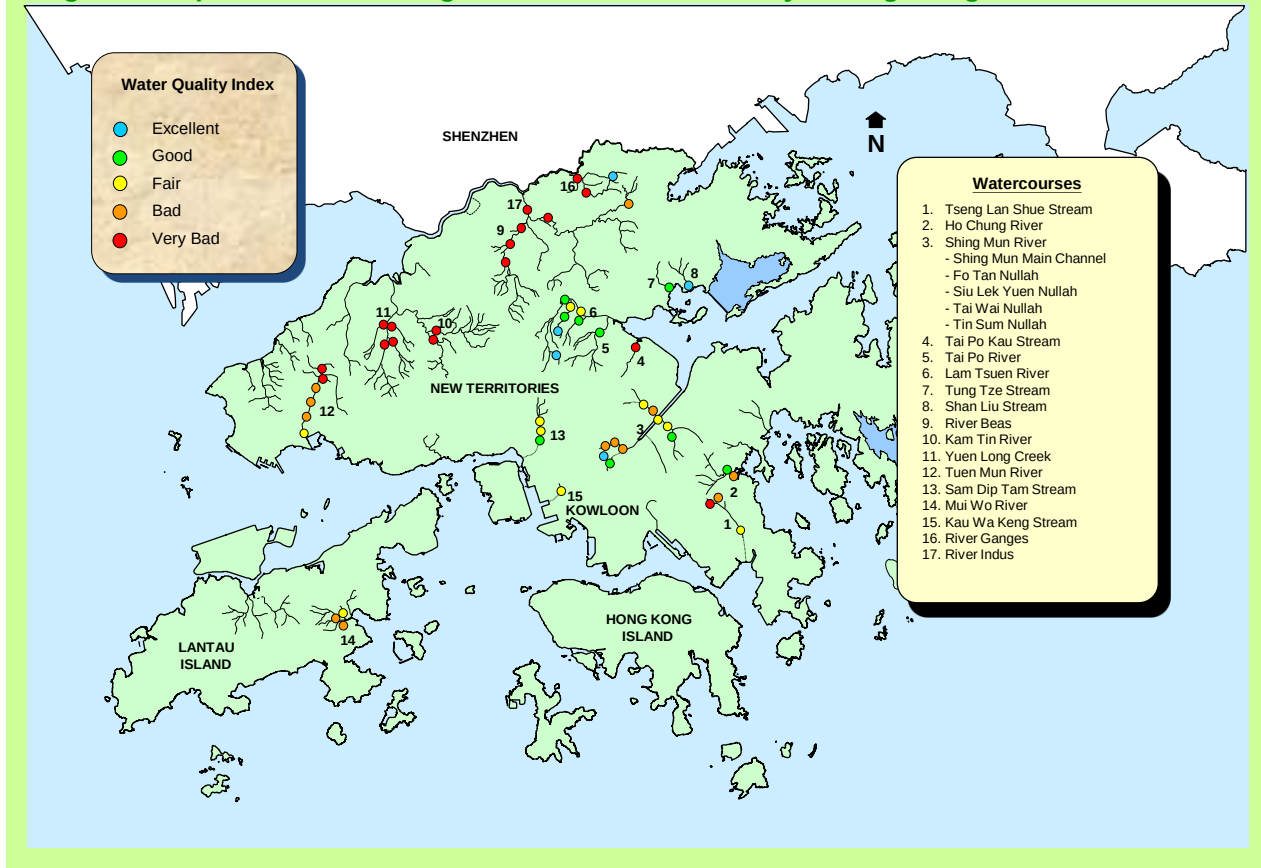


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Figure 4. Map of river monitoring stations and Water Quality Index gradings in 2018



Figure 5. Map of river monitoring stations and Water Quality Index gradings in 1987

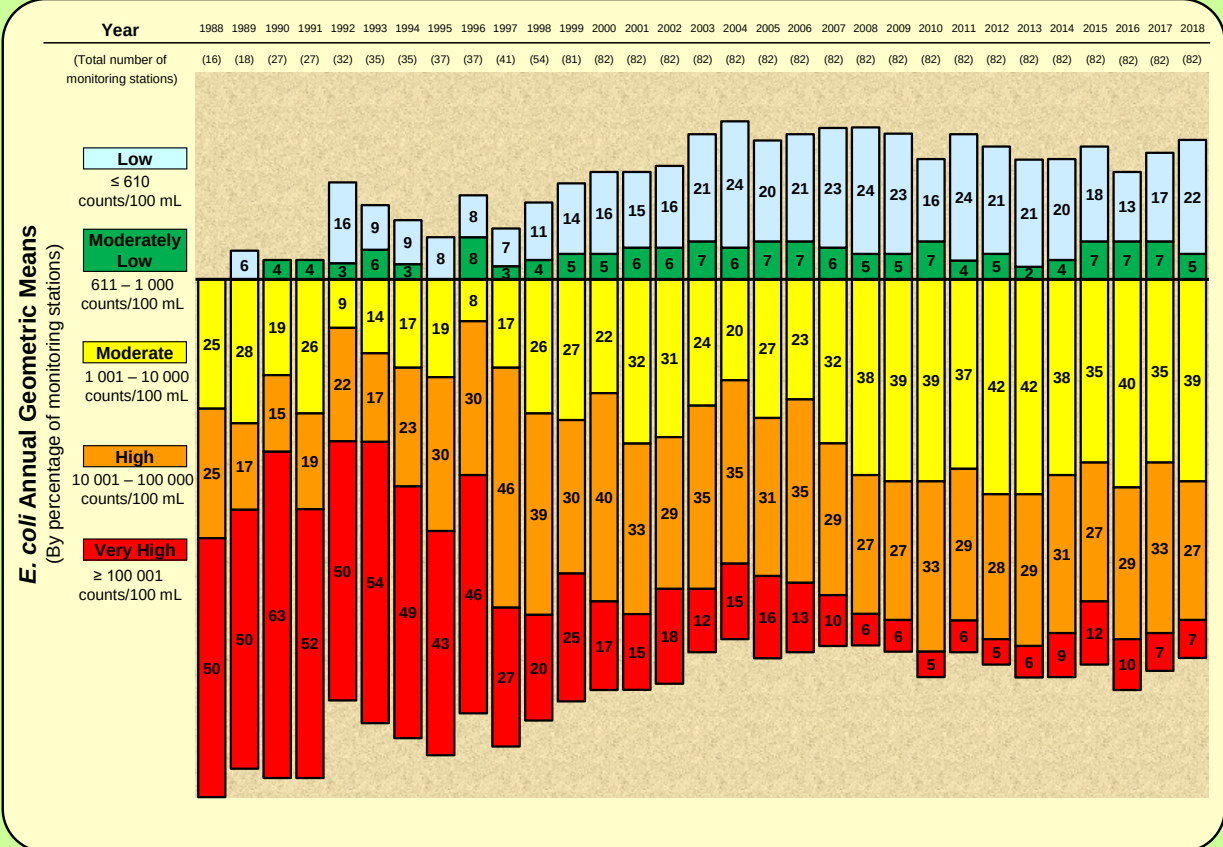


Since 1988, *E. coli* level in river waters was also analysed. As faeces of all warm-blooded animals contain *E. coli*, its measured level in a water body is a common indicator of the presence and extent of faecal contamination. In 2018, 27% of the monitoring stations had “Low” or “Moderately Low” levels of *E. coli* (i.e. less than or equal to 1 000 counts/100mL) while 34% recorded “High” or “Very High” (i.e. over 10 000 counts/100mL) levels² (Figure 6).

Most of the stations with “Very High” levels of *E. coli* are located in the western part of the New Territories (Figure 7), mainly due to discharges from livestock farms, runoff from unsewered village houses as well as expedient connections in old districts. While these watercourses are primarily intended for stormwater conveyance and flood prevention, the Government will continue its efforts in enforcement of pollution control legislations and extension of the public sewer network to further reduce the *E. coli* levels at these locations.

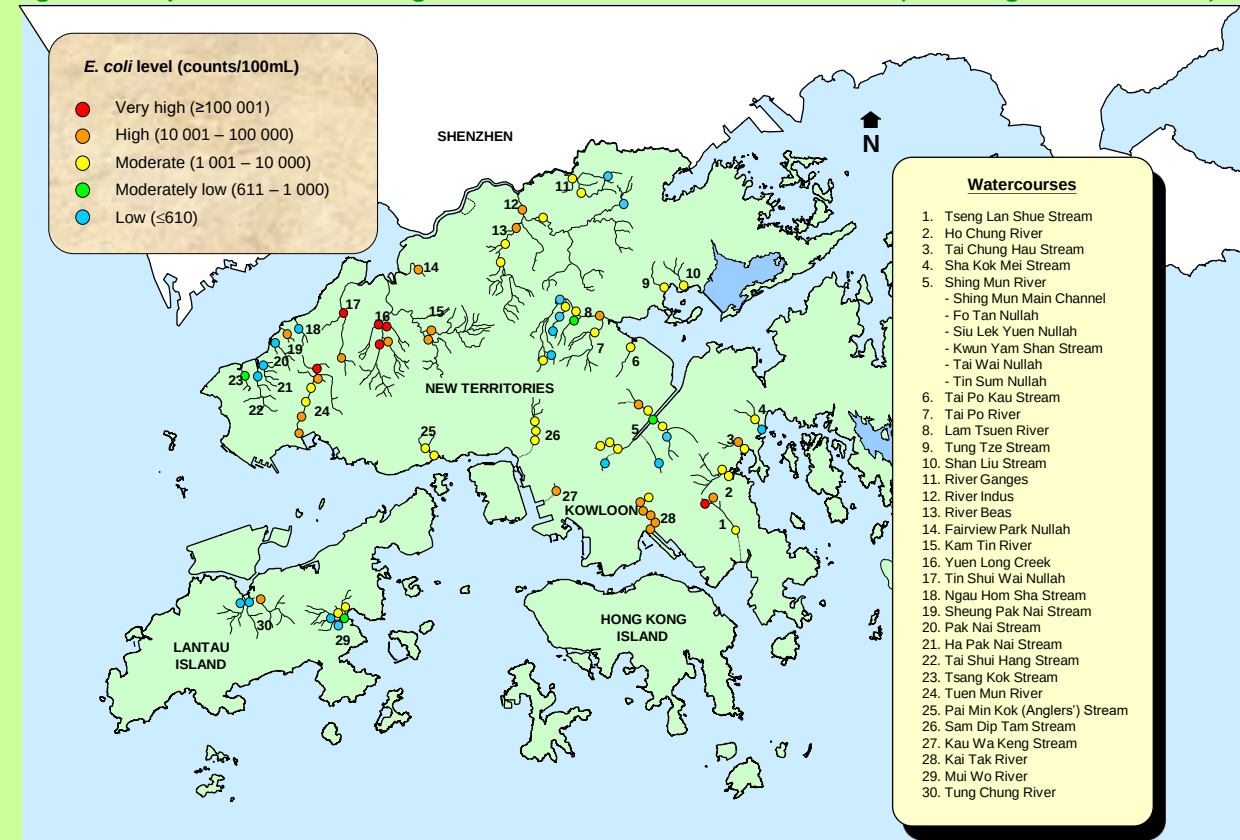
² All levels of *E. coli* in this report are reported as annual geometric means (counts/100mL).

Figure 6. Annual geometric mean *E. coli* levels for river waters of Hong Kong (1988-2018)



Figures are rounded to the nearest integer

Figure 7. Map of river monitoring stations and levels of *E. coli* in 2018 (Annual geometric mean)

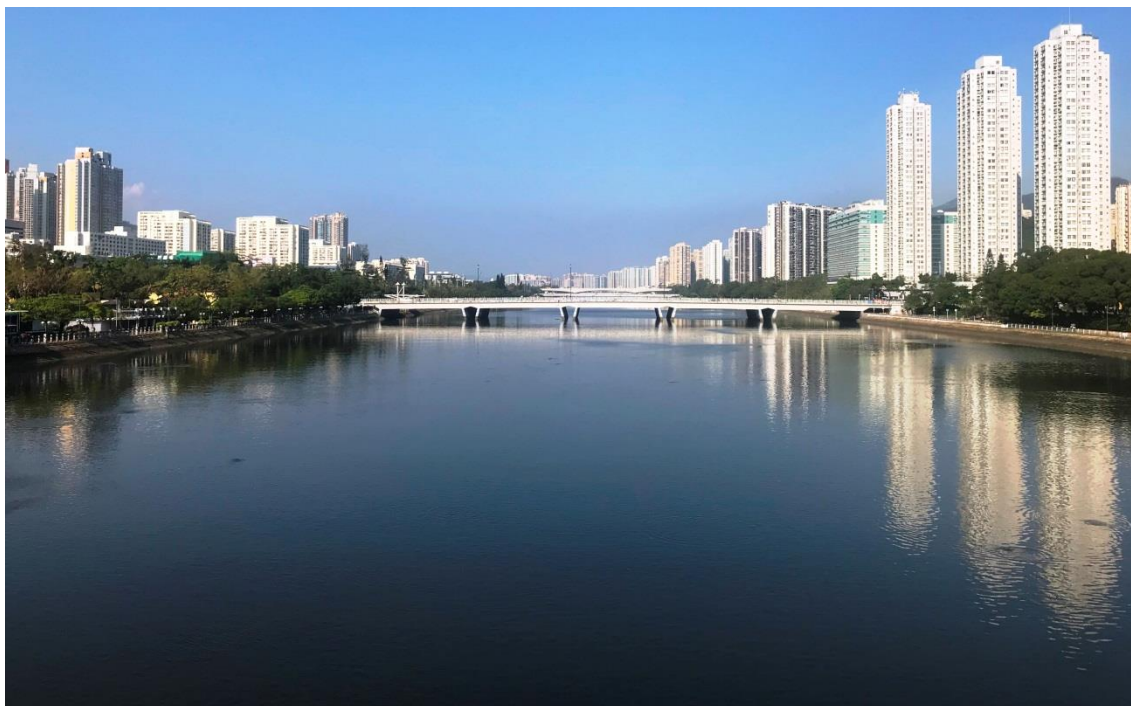


3. Water Quality State at Rivers and Streams

3.1. Eastern New Territories

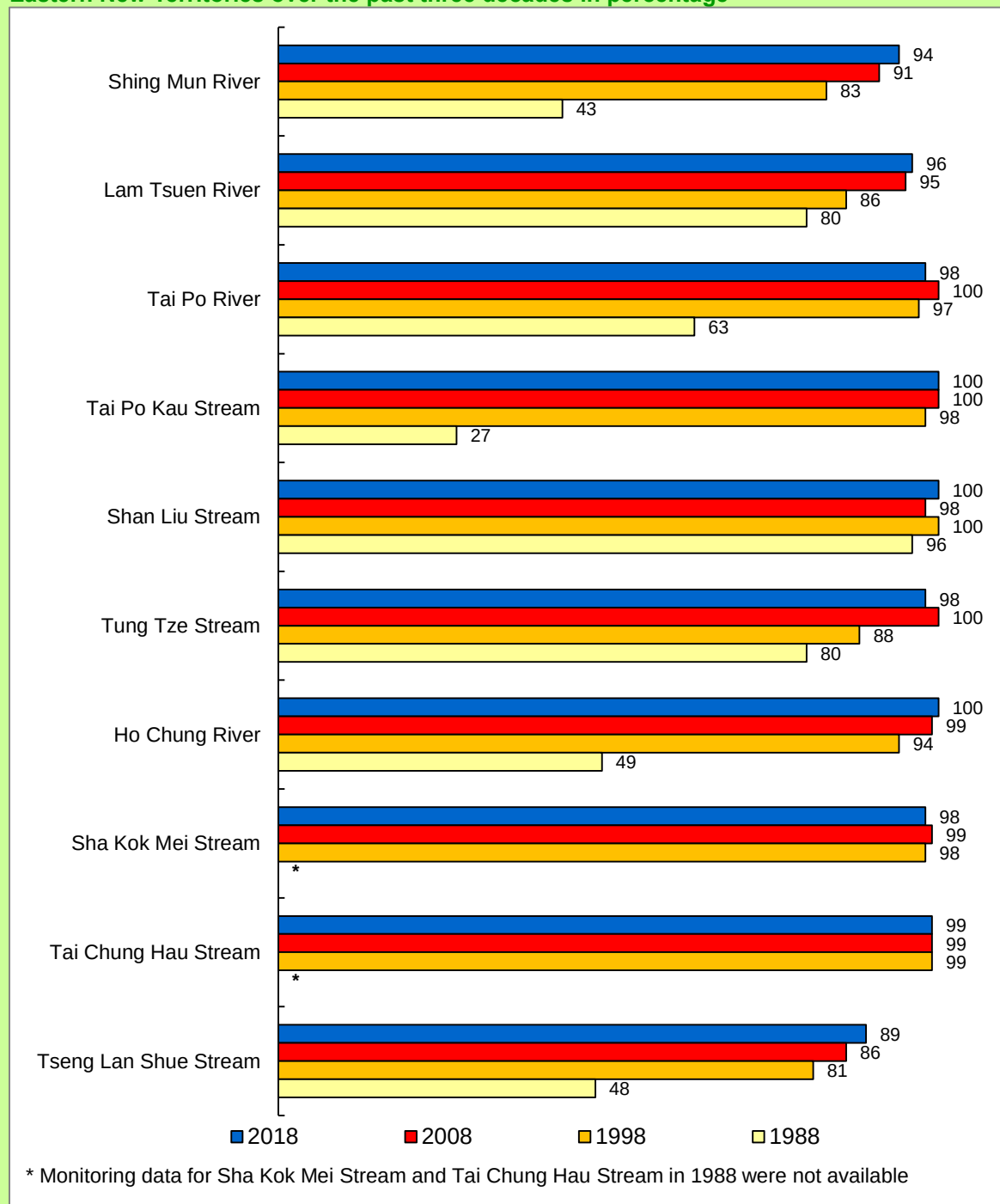
The EPD monitored ten watercourses in the Eastern New Territories in 2018. Six are located in the Tolo Harbour and Channel Water Control Zone (WCZ), including Shing Mun River, Lam Tsuen River, Tai Po River, Tai Po Kau Stream, Shan Liu Stream and Tung Tze Stream. Within the Port Shelter WCZ, there are Ho Chung River, Sha Kok Mei Stream and Tai Chung Hau Stream. Tseng Lan Shue Stream is located within the Junk Bay WCZ.

The water quality of rivers and streams in the Eastern New Territories is good. In 2018, the overall WQO compliance rate of all monitoring stations in the Eastern New Territories was 96%, as compared with 94% in 2008, 88% in 1998 and 58% in 1988 (Figure 8). Within the region, three rivers achieved full compliance (100%) with the WQOs in 2018. These are Tai Po Kau Stream and Shan Liu Stream in the Tolo Harbour and Channel WCZ and Ho Chung River in the Port Shelter WCZ.



Shing Mun River

Figure 8. Compliance with the Water Quality Objectives of the rivers and streams in the Eastern New Territories over the past three decades in percentage



Shing Mun River, a major river which has three main tributaries and runs through the densely populated Sha Tin urban area, showed marked improvement during the past three decades. The WQO compliance rate of Shing Mun River was 94% in 2018. Lam Tsuen River, a major river passing through the urban area of Tai Po which joins Tai Po River before entering Tolo Harbour, recorded a WQO compliance rate of 96% in 2018. Both Tai Po River and Tung Tze Stream achieved an overall WQO compliance rate of 98% in 2018.



Lam Tsuen River

In the Port Shelter WCZ, the WQO compliance for Ho Chung River, Sha Kok Mei Stream and Tai Chung Hau Stream were 100%, 98% and 99% respectively in 2018.

Tseng Lan Shue Stream in the Junk Bay WCZ had a WQO compliance rate of 89% in 2018.

As for the WQI grading, 31 out of 32 (97%) of the river monitoring stations in the Eastern New Territories were graded “Good” or “Excellent” in 2018, same as 2017 (Figures 9 – 13). The only station in the “Bad” category was at the Tseng Lan Shue Stream (JR3) (Figure 12).

The *E. coli* level at Shing Mun River Main Channel (TR19I), the only watercourse currently used for secondary contact recreation activities in Hong Kong, was rated “Moderately low” in 2018, same as 2017.

Figure 9. Water Quality Index gradings and *E. coli* levels in Shing Mun River

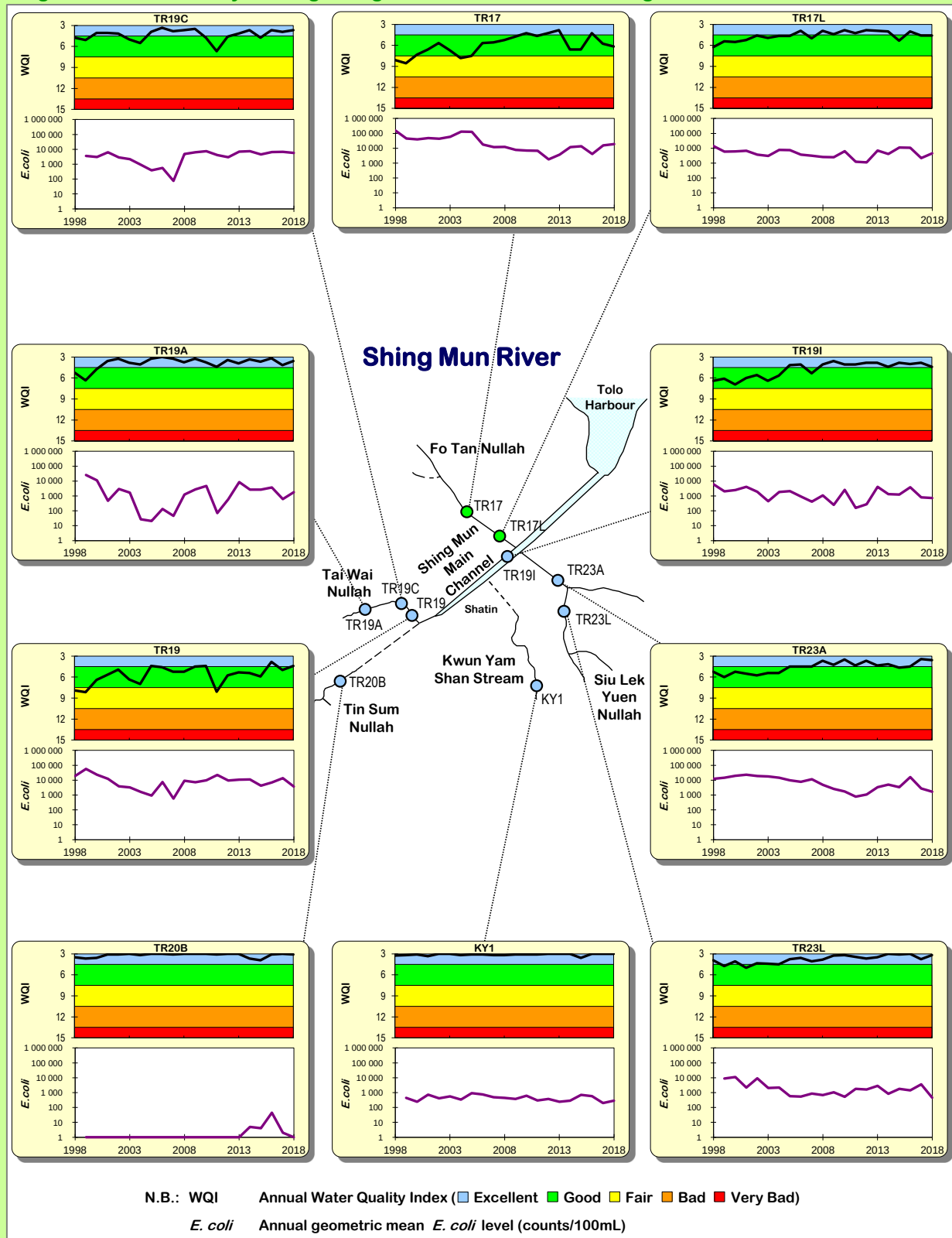


Figure 10. Water Quality Index gradings and *E. coli* levels in Lam Tsuen River and Tai Po River

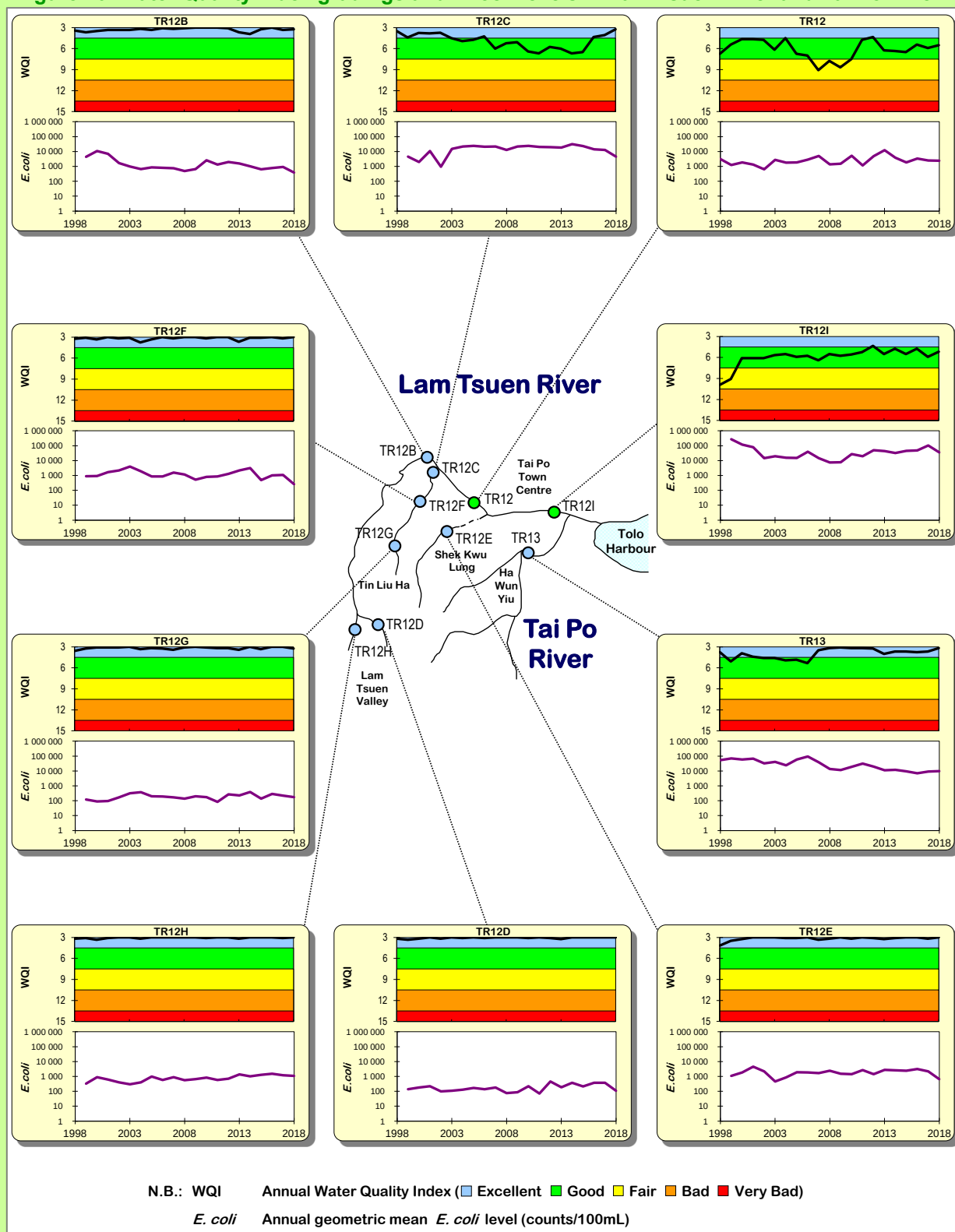


Figure 11. Water Quality Index gradings and *E. coli* levels in Tai Po Kau Stream, Shan Liu Stream and Tung Tze Stream

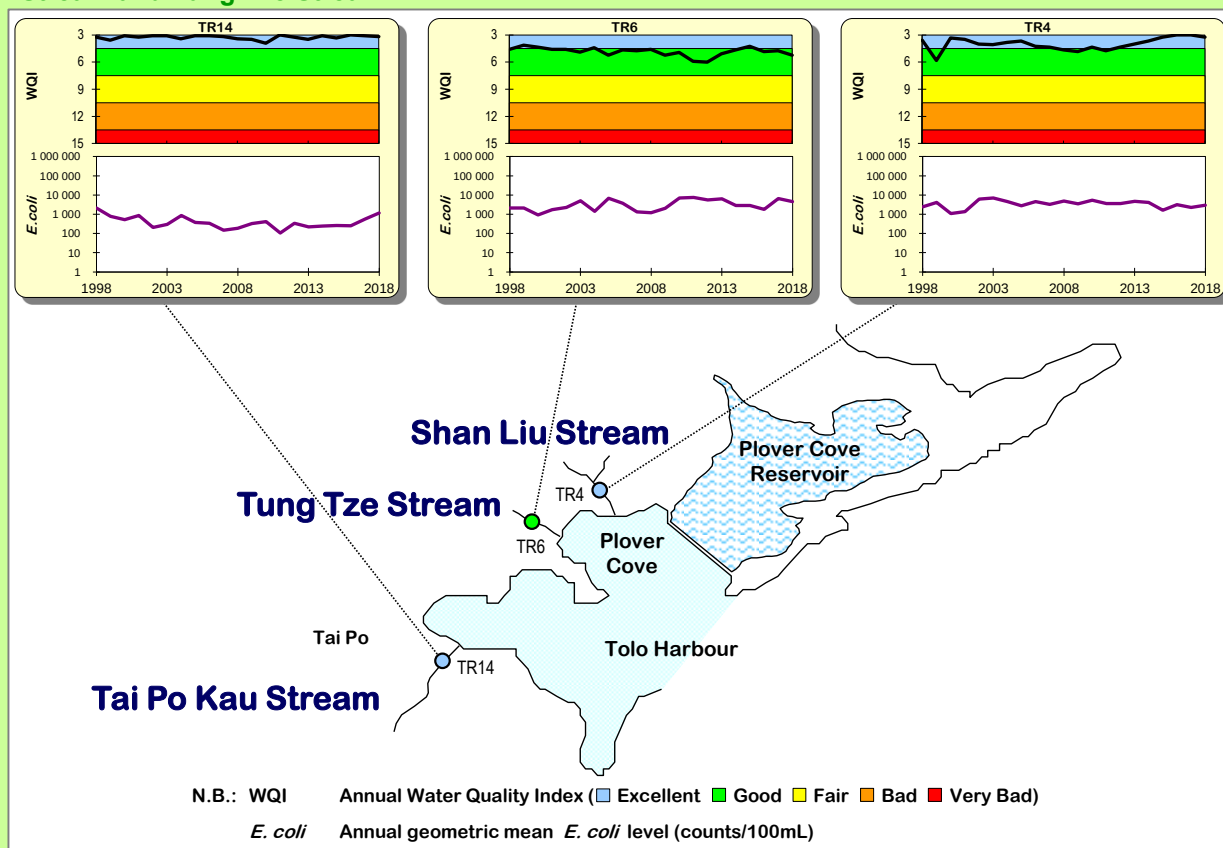


Figure 12. Water Quality Index gradings and *E. coli* levels in Tseng Lan Shue Stream

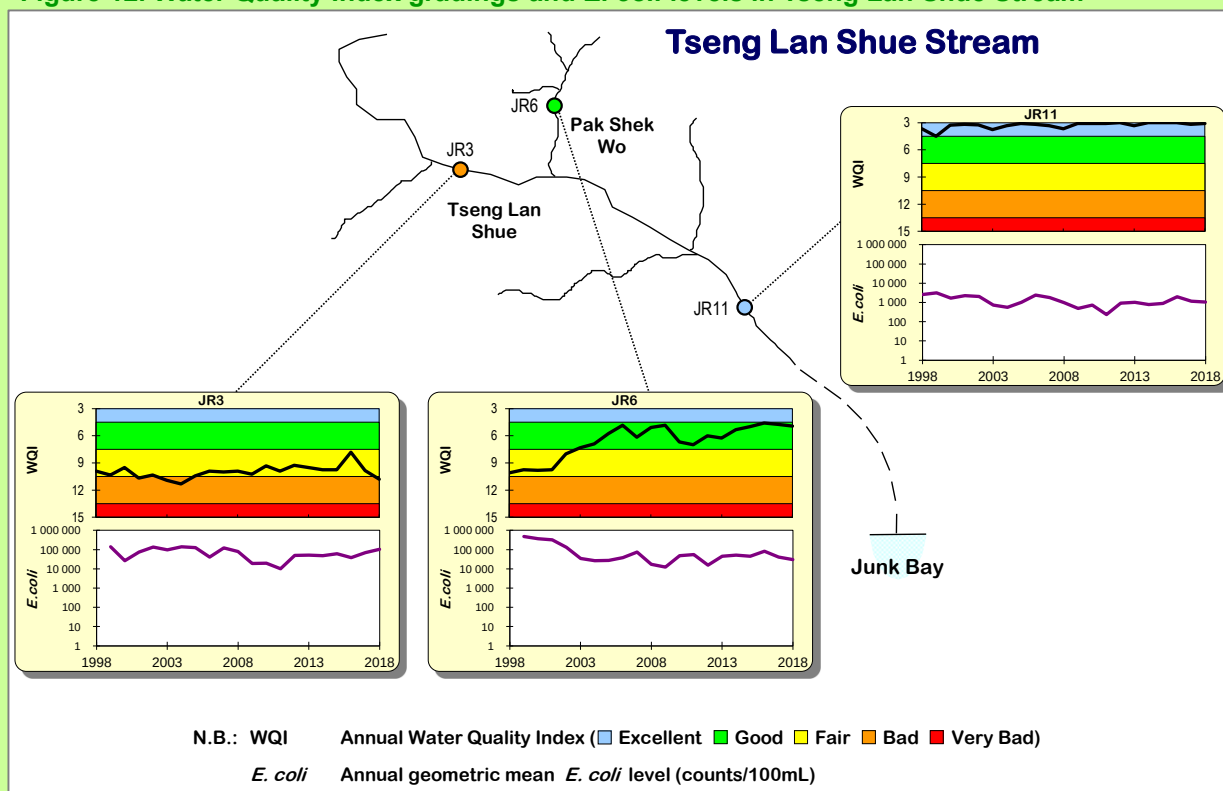
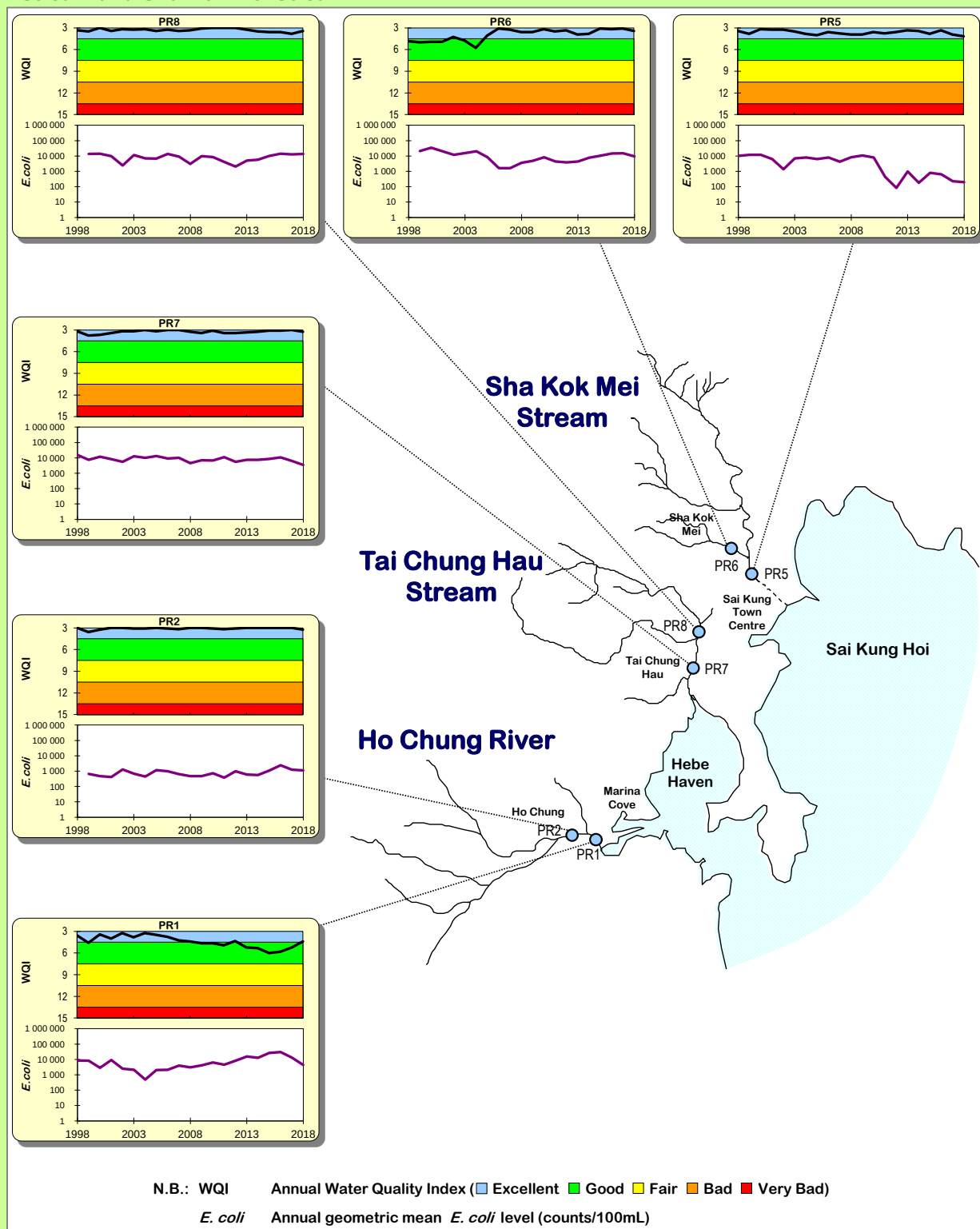


Figure 13. Water Quality Index gradings and *E. coli* levels in Ho Chung River, Tai Chung Hau Stream and Sha Kok Mei Stream



3.2. Northwestern New Territories

In the Northwestern New Territories, the EPD monitors a total of 13 rivers and streams, which flow into Shenzhen River or directly into Deep Bay (Shenzhen Bay). These include River Indus, Beas and Ganges in the North District, Yuen Long Creek, Kam Tin River, Tin Shui Wai Nullah and Fairview Park Nullah in the Yuen Long District, and the six smaller streams located around Lau Fau Shan area, all within the Deep Bay WCZ.



River Indus

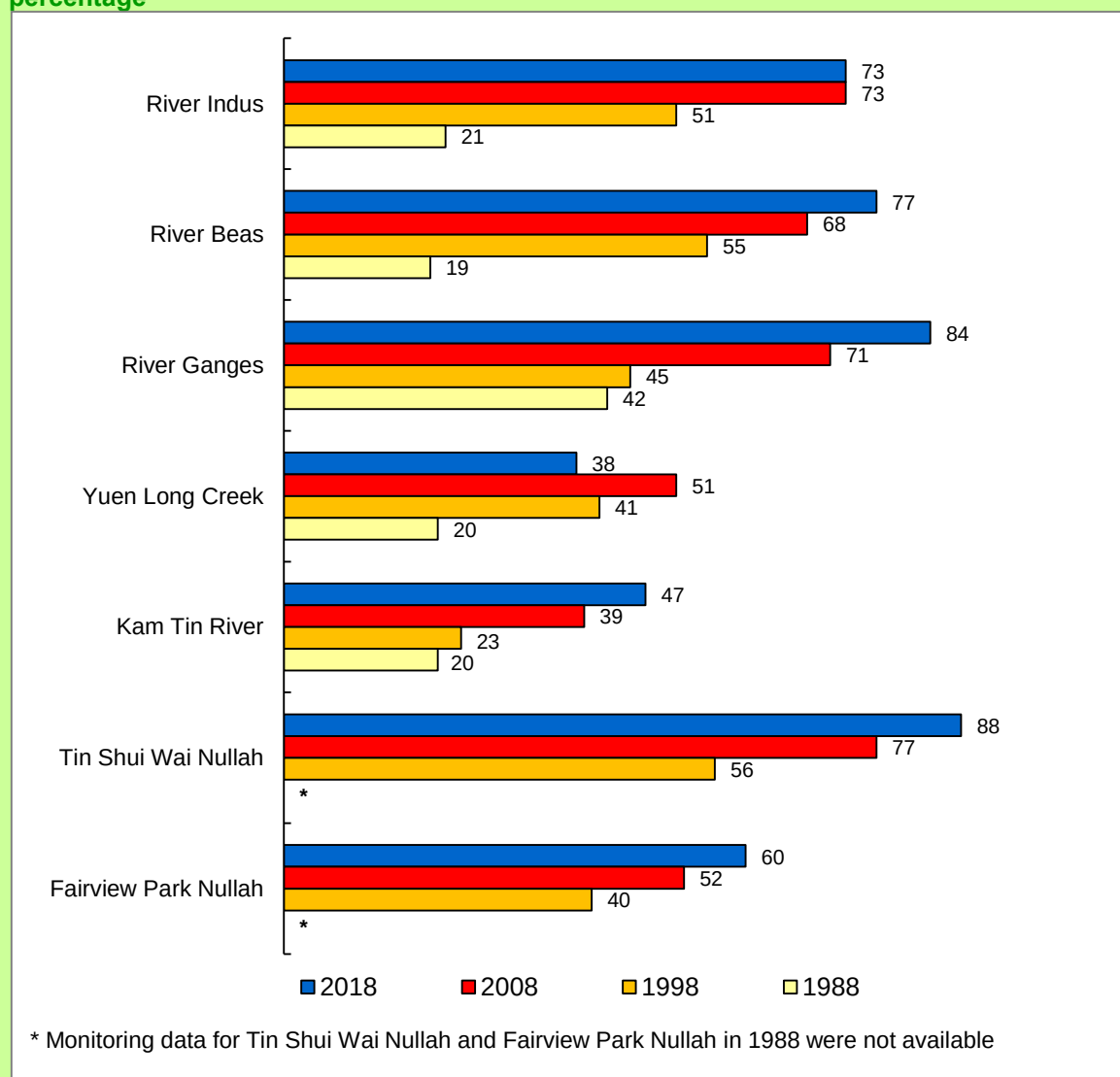
Water quality of rivers and streams in the Northwestern New Territories showed improvements over the past three decades. In 2018, the overall WQO compliance rate of all monitoring stations in the Northwestern New Territories was 74%, as compared with 72% in 2008, 60% in 1998 and 24% in 1988 (Figure 14).

River Indus is a major river in the North District. It runs through rural areas like Lung Yeuk Tau, collects runoff from the densely populated Fanling and Sheung Shui urban areas, meets with River Beas before draining into Shenzhen River. The overall WQO compliance rate was 73% in 2018, as compared with 21% in 1988. In 2018, its downstream station (IN1) had a “Fair” grading while the mid-stream (IN2) and upstream (IN3) stations were “Good” and “Excellent” respectively, same as 2017 (Figure 16).

As a tributary of River Indus, River Beas recorded an overall WQO compliance rate of 77% in 2018, as compared with 19% in 1988. Individually, the upstream, mid-stream and downstream stations (RB1, RB2 and RB3) received “Good”, “Good” and “Fair” grading respectively in 2018 (Figure 16).

For River Ganges, the overall WQO compliance rate was 84% in 2018, as compared with 42% in 1988. In terms of WQI, both downstream station (GR1) and mid-stream station (GR2) were graded “Fair”, while the upstream station (GR3) was graded “Excellent” in 2018 (Figure 16).

Figure 14. Compliance with the Water Quality Objectives of the rivers and streams in the Northwestern New Territories (North District and Yuen Long) over the past three decades in percentage



As a major river in Yuen Long District, Yuen Long Creek passes through rural areas as well as the densely populated Yuen Long new town and Yuen Long Kau Hui, joins Kam Tin River before entering Deep Bay. In 2018, Yuen Long Creek’s overall WQO compliance rate was 38%, as

compared with 20% in 1988. For Kam Tin River, the overall WQO compliance rate in 2018 was 47%.

In terms of WQI, the two upstream stations (YL1 and YL2) were “Bad”, while the two downstream stations (YL3 and YL4) were “Very Bad” in 2018 due to foul sewer pollution incidents in the catchment areas (Figure 17). For Kam Tin River, one station (KT1) was upgraded to “Fair” in 2018 from “Bad” in 2017. The other station (KT2) was graded as “Bad” in 2018, same as 2017.

To improve the water quality in the above areas, in addition to rectification of defective sewers, the Government is now carrying out a review of the local sewerage programme with a view to reducing the residual pollution load by unsewered villages. Options include the provision of sewerage facilities for unsewered villages and possible installation of dry weather flow interceptors (DWFIs) where appropriate. In addition, EPD has been stepping up enforcement effort against illegal discharges and also collaborating with the Agriculture, Fisheries and Conservation Department in undertaking education programmes to tackle the pollution problem arising from livestock farms which is another key pollution source in the area.



Tin Shui Wai Nullah

Tin Shui Wai Nullah had an overall WQO compliance of 88% in 2018 as compared with 77% in 2008 and 56% in 1998. In terms of WQI, its downstream station (TSR1) was graded “Fair” and its upstream station (TSR2) was graded “Good” in 2018 (Figure 17), same as 2017.

The Fairview Park Nullah station (FVR1) recorded 60% WQO compliance in 2018 as compared with 52% in 2008 and 40% in 1998. The WQI was rated “Fair” in 2018 (Figure 17), same as 2017.

In 2018, the water quality of the six smaller streams around the Lau Fau Shan area all remained good. Ngau Hom Sha Stream achieved full compliance (100%) with the WQOs while the remaining streams recorded an overall WQO compliance of 93% or above (Figure 15). All streams had “Excellent” WQI gradings (Figure 18).

Figure 15. Compliance with the Water Quality Objectives of the smaller streams in Northwestern New Territories (Lau Fau Shan area) over the past two decades in percentage

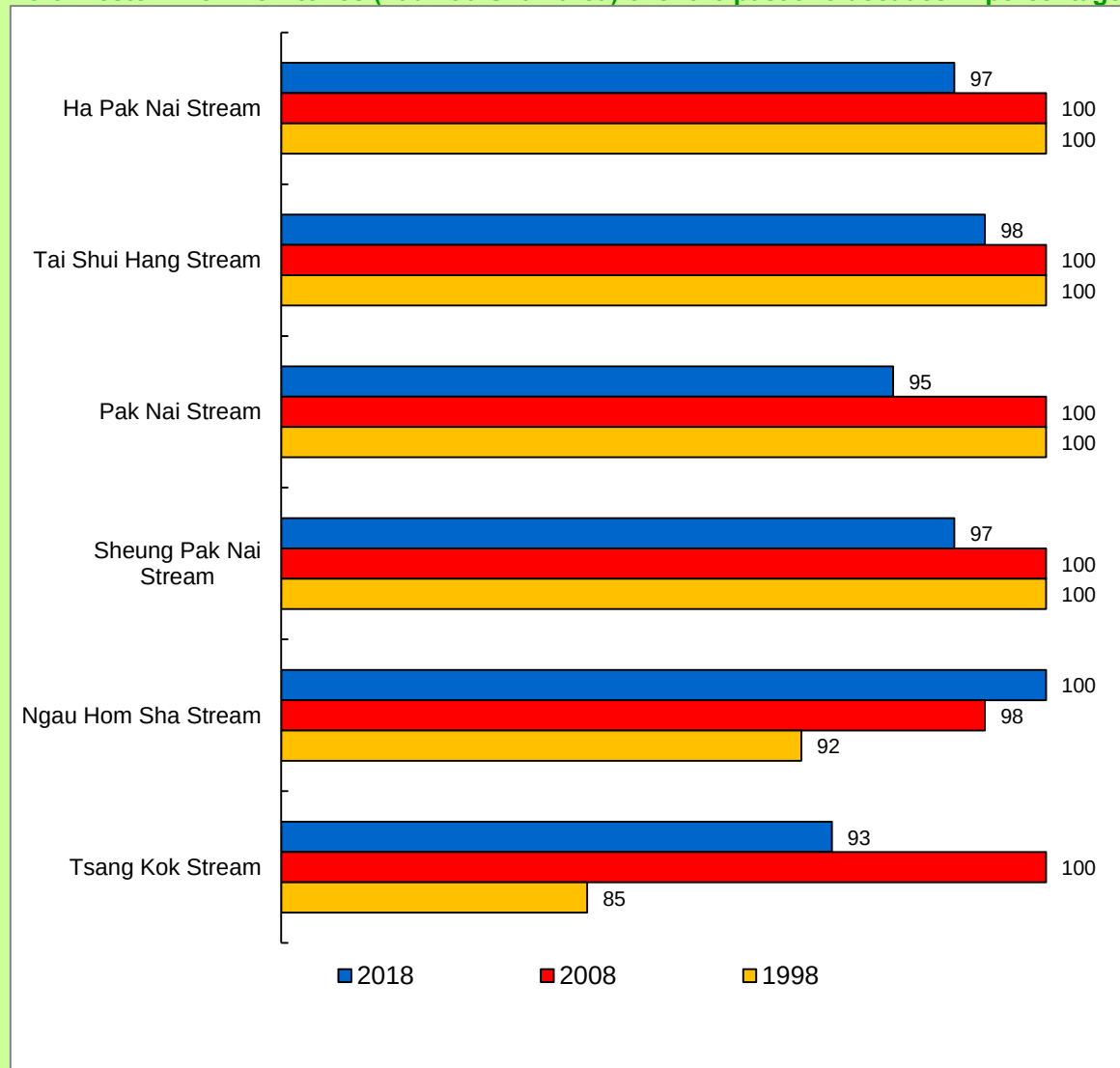


Figure 16. Water Quality Index gradings and *E. coli* levels in River Indus, River Beas and River Ganges

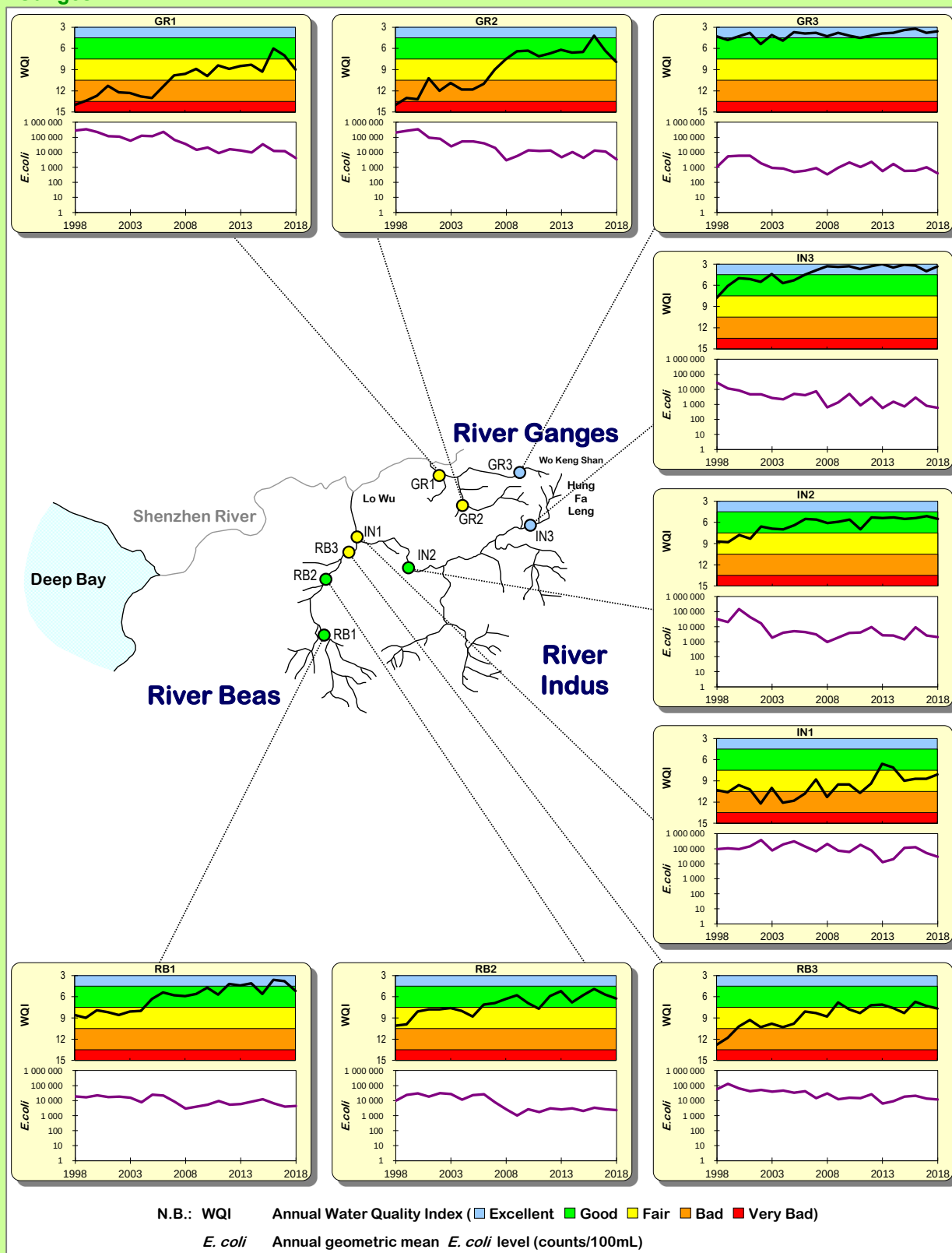


Figure 17. Water Quality Index gradings and *E. coli* levels in Yuen Long Creek, Kam Tin River, Tin Shui Wai Nullah and Fairview Park Nullah

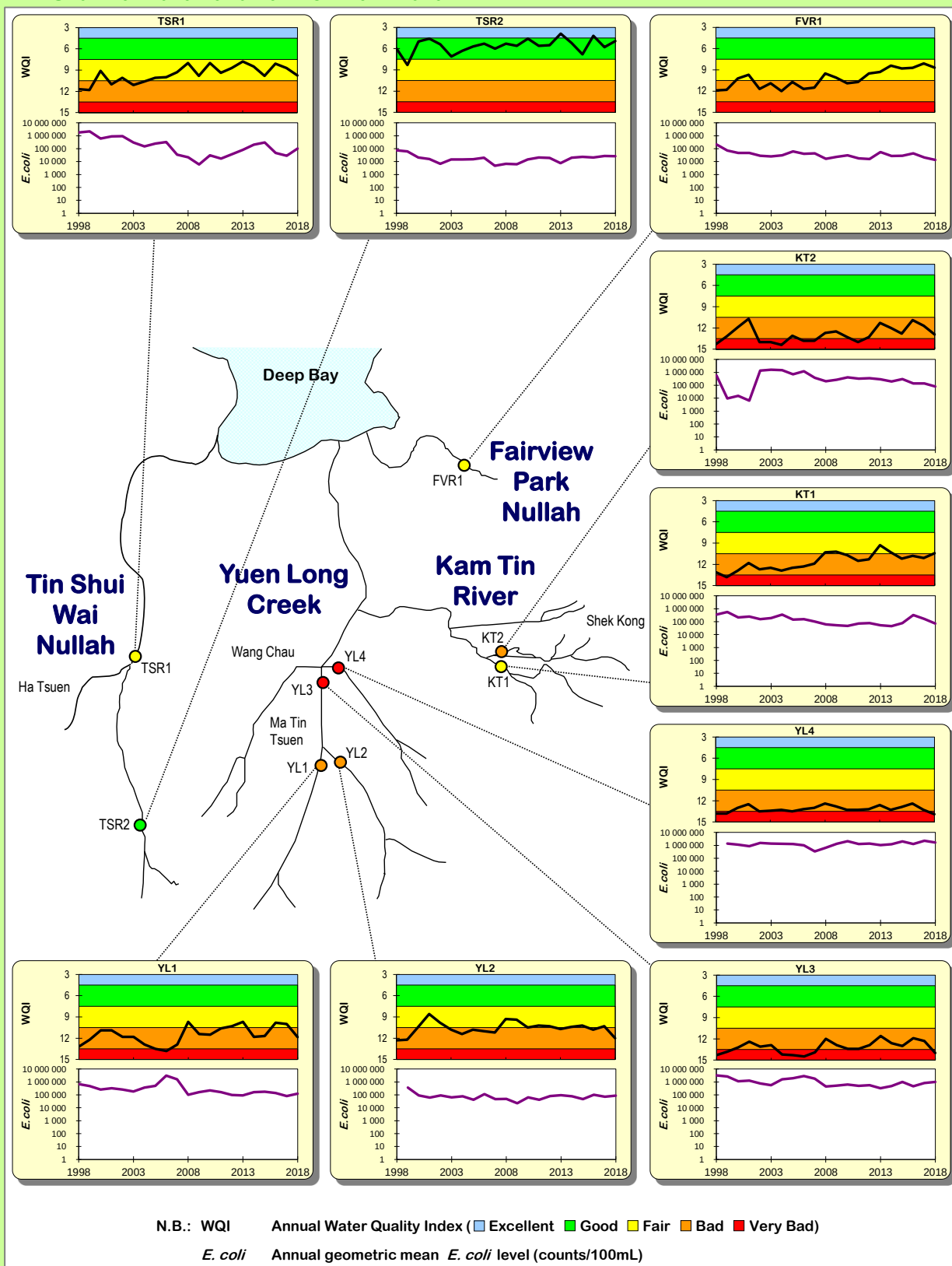
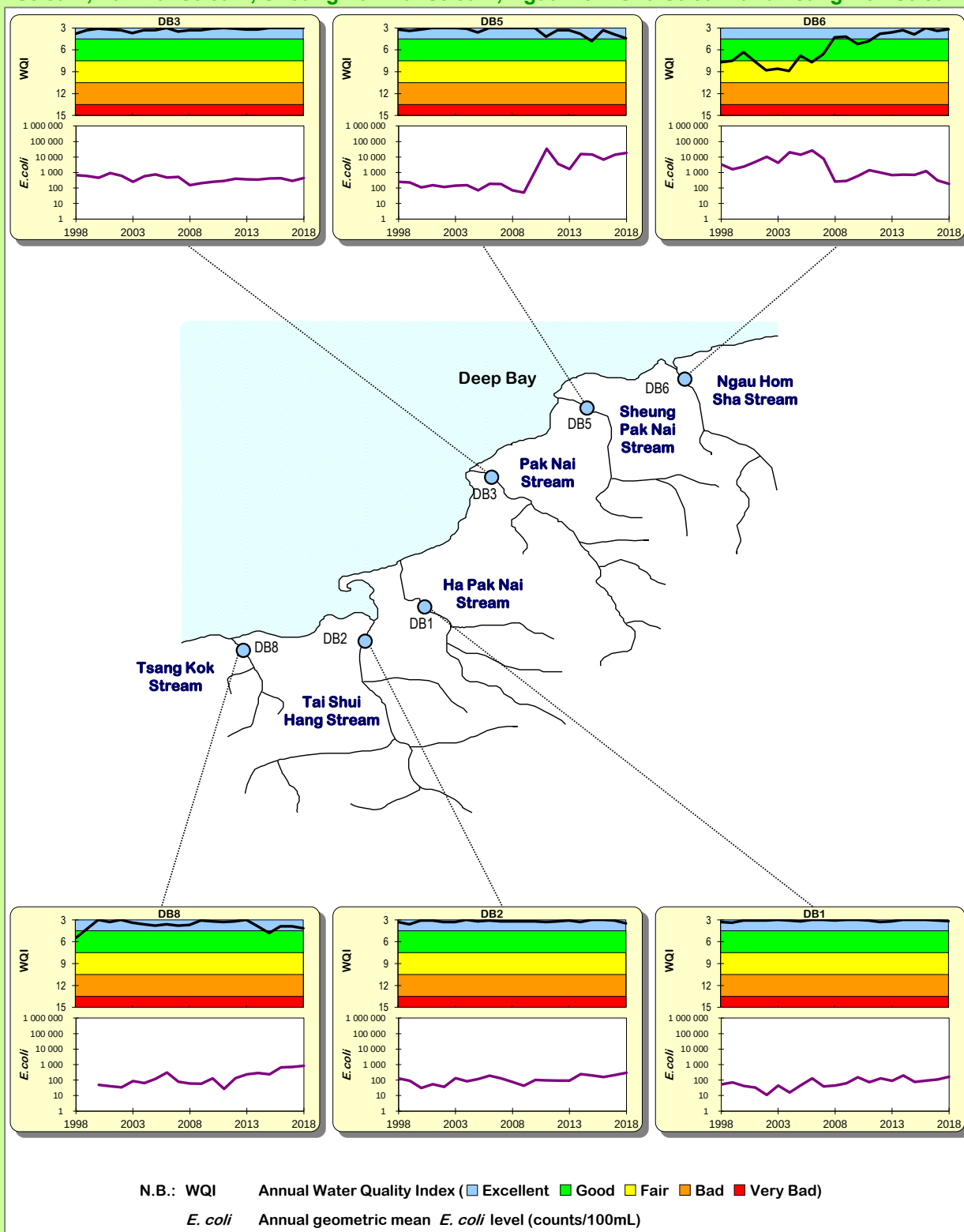


Figure 18. Water Quality Index gradings and *E. coli* levels in Ha Pak Nai Stream, Tai Shui Hang Stream, Pak Nai Stream, Sheung Pak Nai Stream, Ngau Hom Sha Stream and Tsang Kok Stream

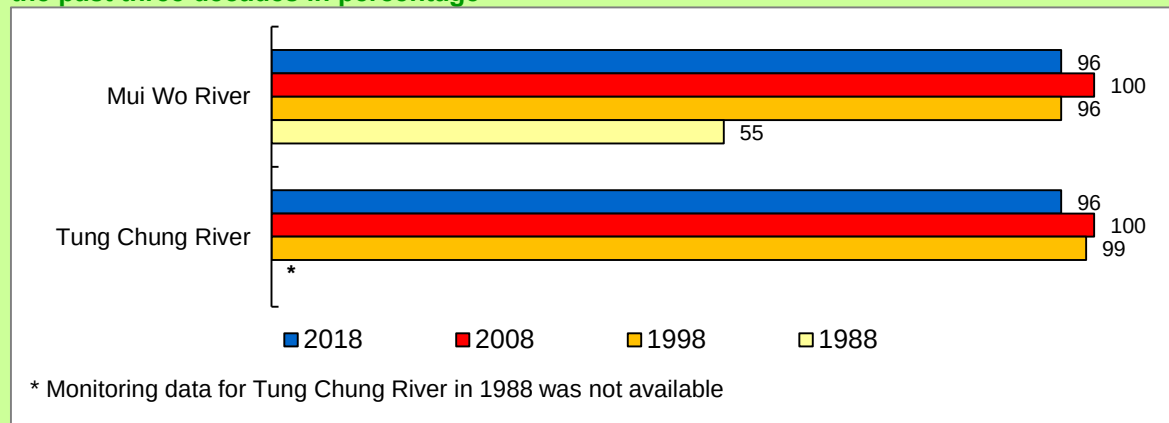


3.3. Lantau Island

Lantau Island is large and less populated. Two rivers with a total of eight monitoring stations are monitored by the EPD: five along Mui Wo River on the south-eastern side of Lantau Island (the Southern WCZ) and three at Tung Chung River on the north-western side of the island (the North Western WCZ).

Mui Wo River and Tung Chung River generally displayed satisfactory water quality. The WQO compliance rate of both rivers in 2018 was 96% (Figure 19). As for the WQI grading, one monitoring station at Mui Wo River (MW5) and another at Tung Chung River (TC3) were graded “Good” (Figures 20 & 21). All of the remaining stations at both rivers were graded “Excellent” in 2018.

Figure 19. Compliance with the Water Quality Objectives of the rivers in Lantau Island over the past three decades in percentage



Tung Chung River

Figure 20. Water Quality Index gradings and *E. coli* levels in Mui Wo River

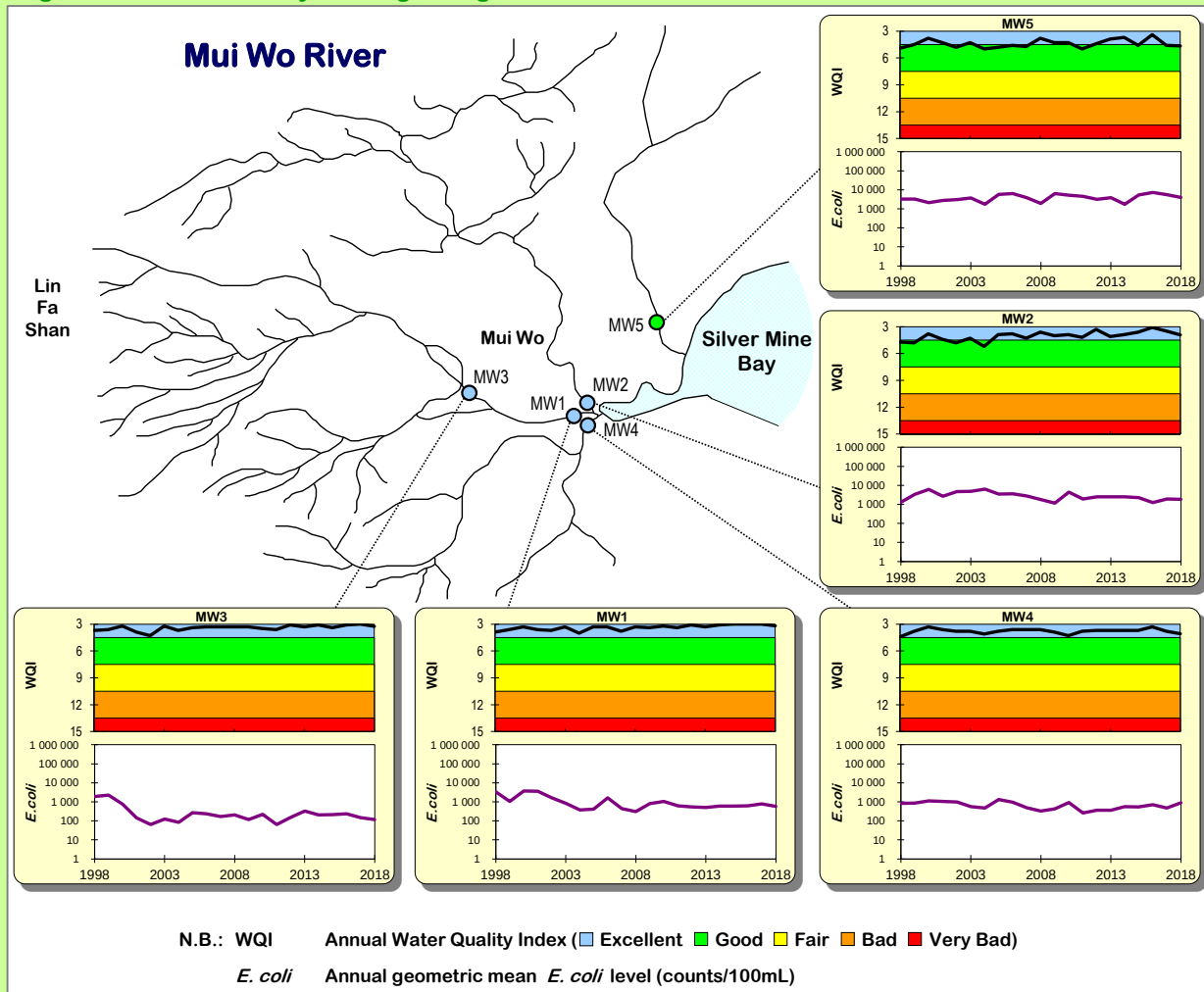
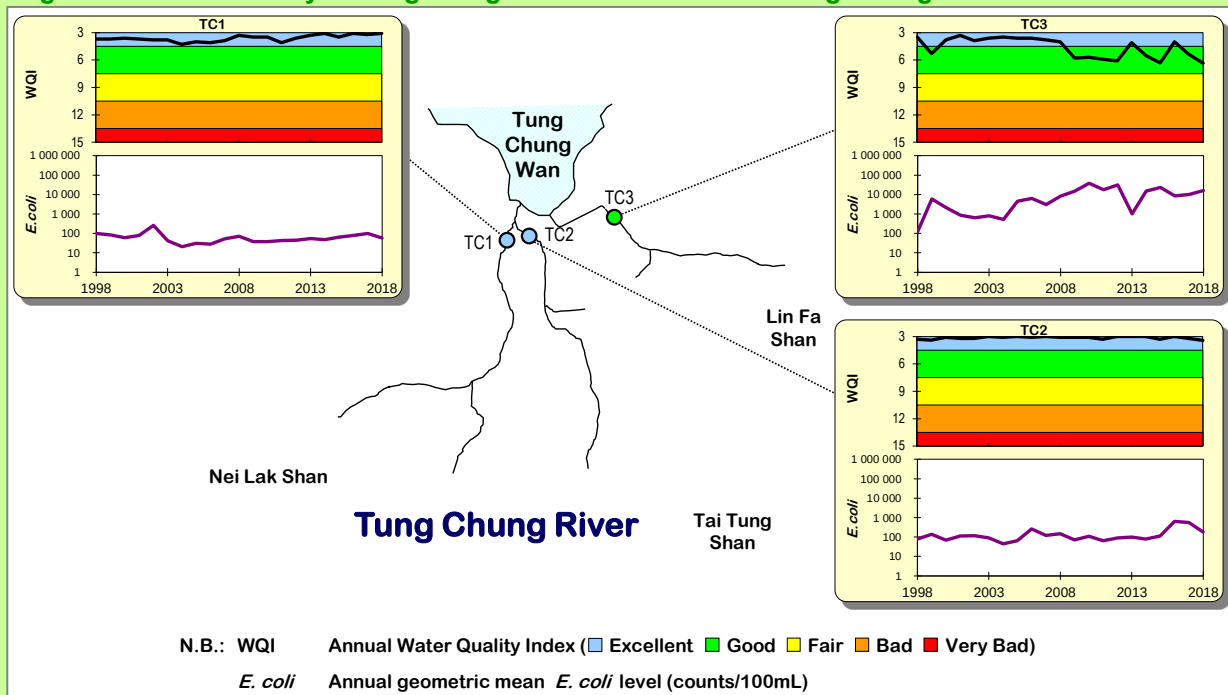
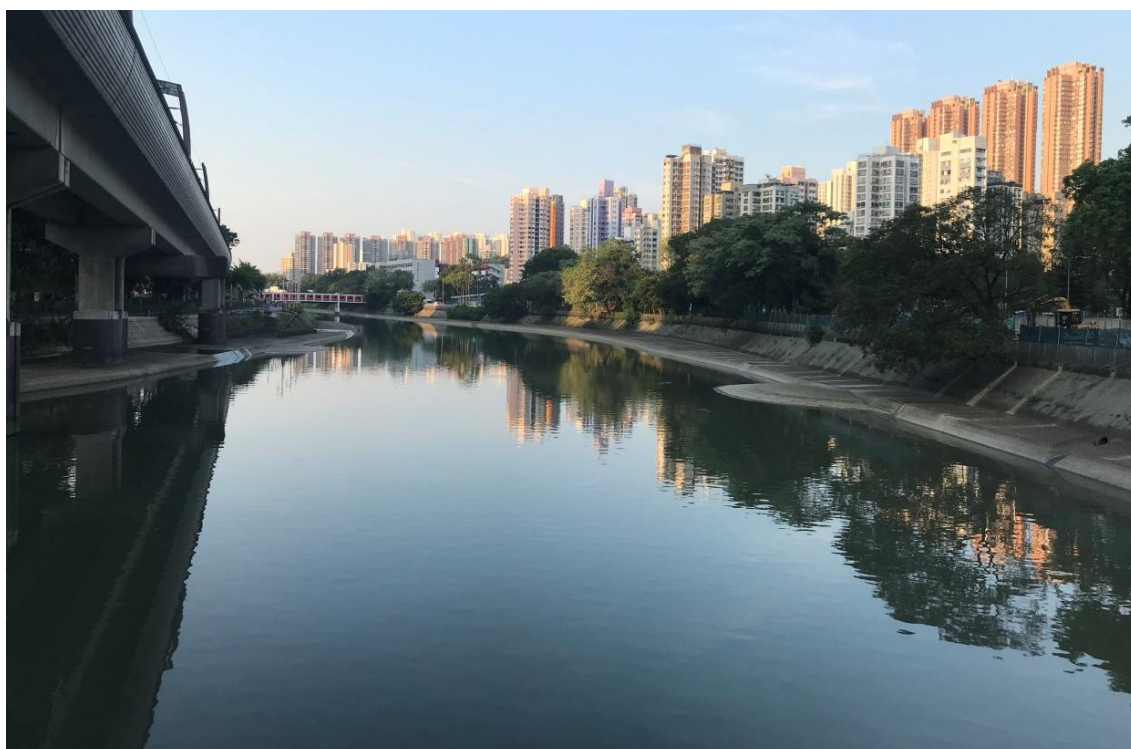


Figure 21. Water Quality Index gradings and *E. coli* levels in Tung Chung River



3.4. Southwestern New Territories and Kowloon

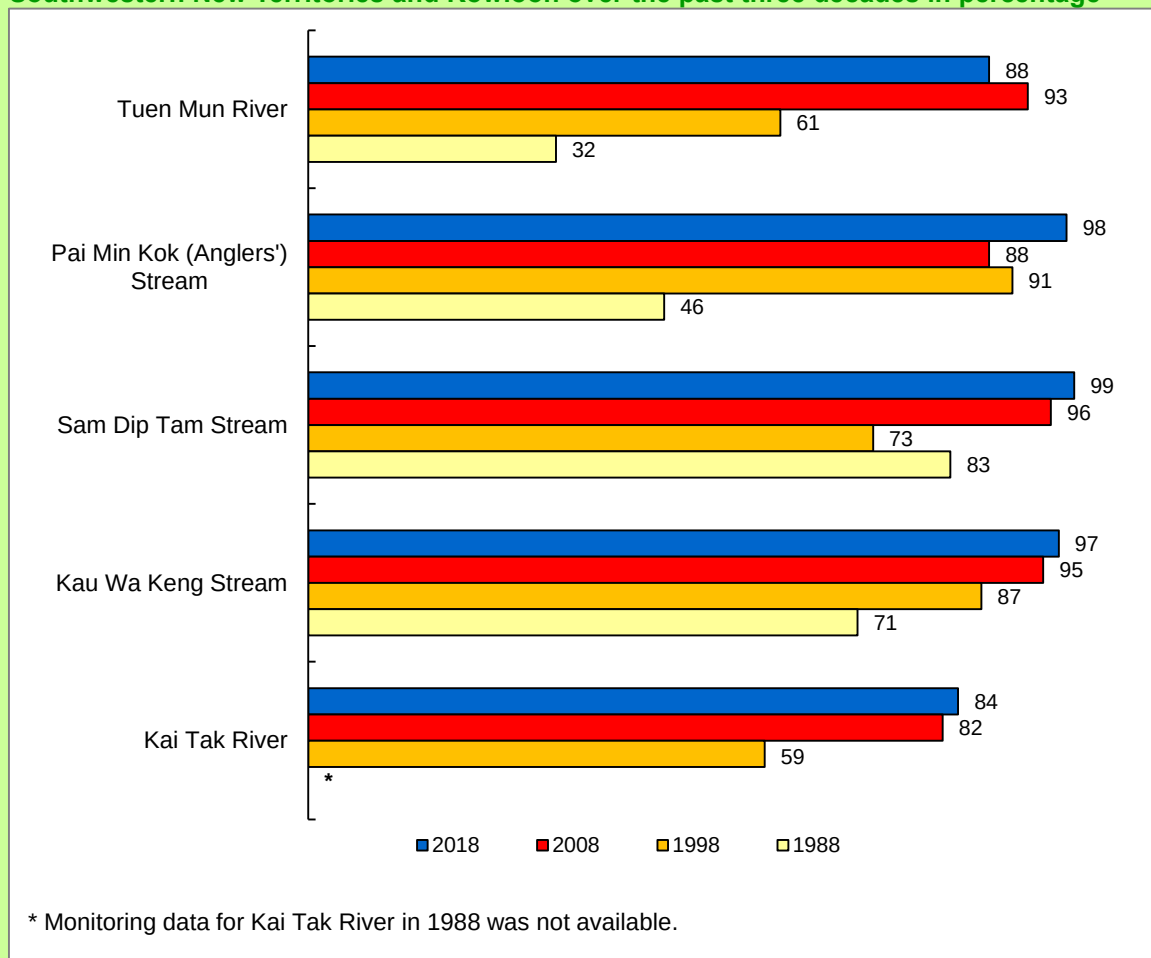
The Southwestern New Territories and Kowloon area stretches from Tuen Mun to the eastern end of Victoria Harbour. The five watercourses being monitored are Tuen Mun River in Tuen Mun (the North Western WCZ), Pai Min Kok Stream near Sham Tseng (the Western Buffer WCZ), Sam Dip Tam Stream near Tsuen Wan, Kau Wa Keng Stream in Kwai Chung and Kai Tak River located in Kowloon (the Victoria Harbour WCZ). There was substantial improvement in the water quality of these urban watercourses over the past 30 years (Figure 22).



Tuen Mun River

In 2018, 17 out of the 18 river monitoring stations (94%) in the Southwestern New Territories and Kowloon were graded “Excellent” or “Good” (Figures 23 – 25), same as 2017. Among these stations, only one station was graded “Bad” this year (i.e. TN1 at Tuen Mun River upstream), as in 2017 and 2016. The overall WQO compliance rate of the rivers and streams in the area has increased from 67% in 1998 to 90% in 2018.

Figure 22. Compliance with the Water Quality Objectives of the rivers and streams in the Southwestern New Territories and Kowloon over the past three decades in percentage



As a major river in the Southwestern New Territories, Tuen Mun River passes through Lam Tei, San Hing Tsuen and Fu Tei in its upstream section, and then the densely populated Tuen Mun town in its mid-stream before entering the Tuen Mun Typhoon Shelter. It showed marked improvement in the last three decades, with its WQO compliance rate rising steadily from 32% in 1988 to 88% in 2018. This improvement was attributed to pollution control efforts as well as implementation of the mitigation measures recommended under the Tuen Mun Sewerage Master Plan (SMP). In 2018, the upstream station (TN1) was graded “Bad” mainly due to discharges from unsewered areas (Figure 23). To prevent the pollutants from affecting Tuen Mun River, a DWFI has been installed at West Rail Siu Hong Station to divert the flow at TN1 to the foul sewers and sewage treatment works for treatment. All other five stations (TN2, TN3, TN4, TN5 and TN6) were graded “Good”.

Pai Min Kok Stream achieved an overall WQO compliance rate of 98% in 2018. In terms of WQI, both its upstream station (AN2) and downstream station (AN1) were graded “Excellent” in 2018 (Figure 24), same as 2017.

Sam Dip Tam Stream of Tsuen Wan achieved 99% compliance with the WQO and all its three monitoring stations were graded “Excellent” in 2018 (Figure 24), same as 2017.

The monitoring station (KW3) at Kau Wa Keng Stream in Kwai Chung achieved a WQO compliance rate of 97% in 2018, as compared with 71% in 1988. Its WQI grading was “Excellent” in 2018 (Figure 24), same as 2017.

Kai Tak River had a WQO compliance rate of 84% in 2018, as compared with 59% in 1998. All six monitoring stations in Kai Tak River were grade “Good” in 2018 (Figure 25), same as 2017. It was expected that the water quality of Kai Tak River would further improve once Civil Engineering and Development Department has completed its improvement work in the Kai Tak Development Area and Drainage Services Department finished the sewer system repairment and upgrading works in its catchment area.



Kai Tak River

Figure 23. Water Quality Index gradings and *E. coli* levels in Tuen Mun River

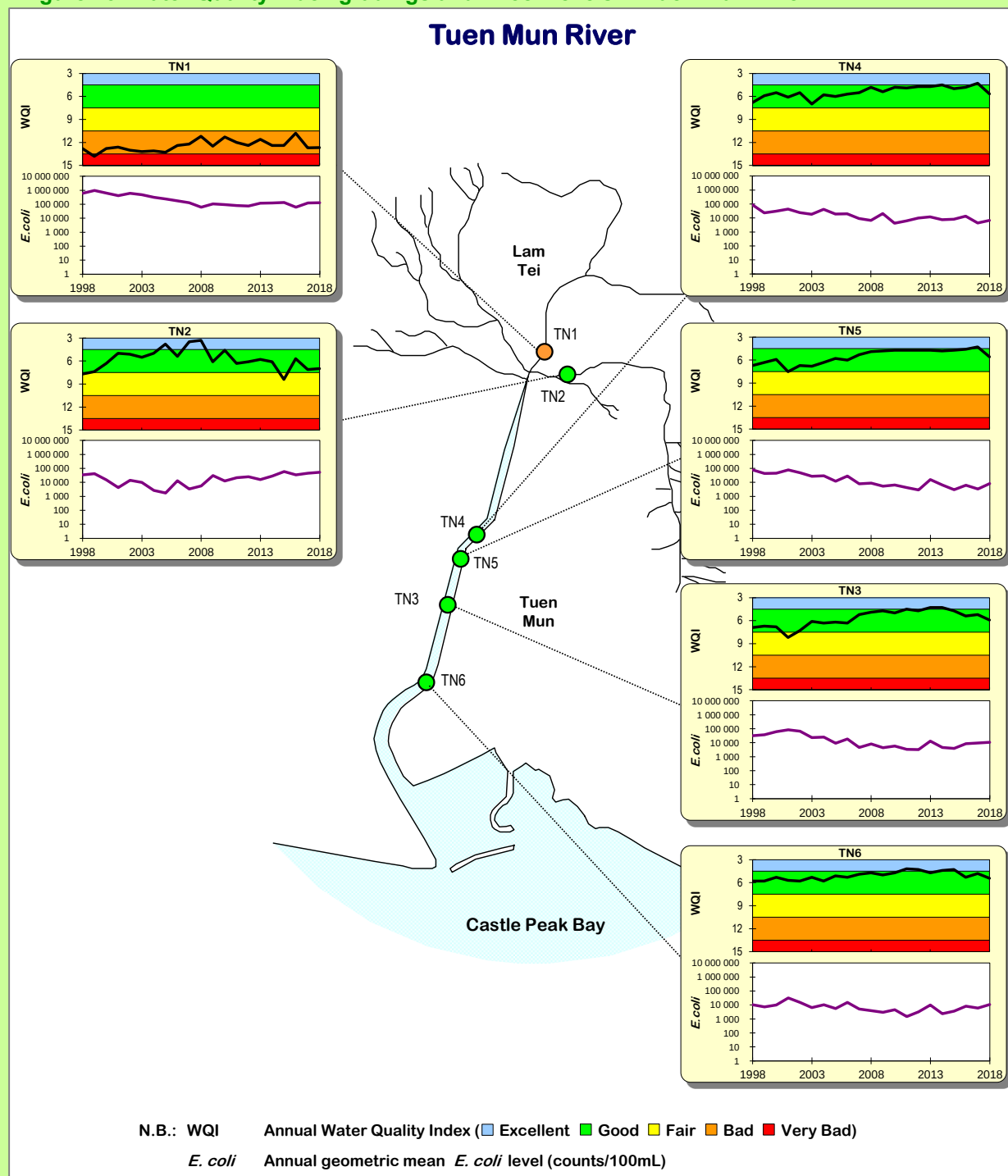


Figure 24. Water Quality Index gradings and *E. coli* levels in Pai Min Kok (Anglers') Stream, Sam Dip Tam Stream and Kau Wa Keng Stream

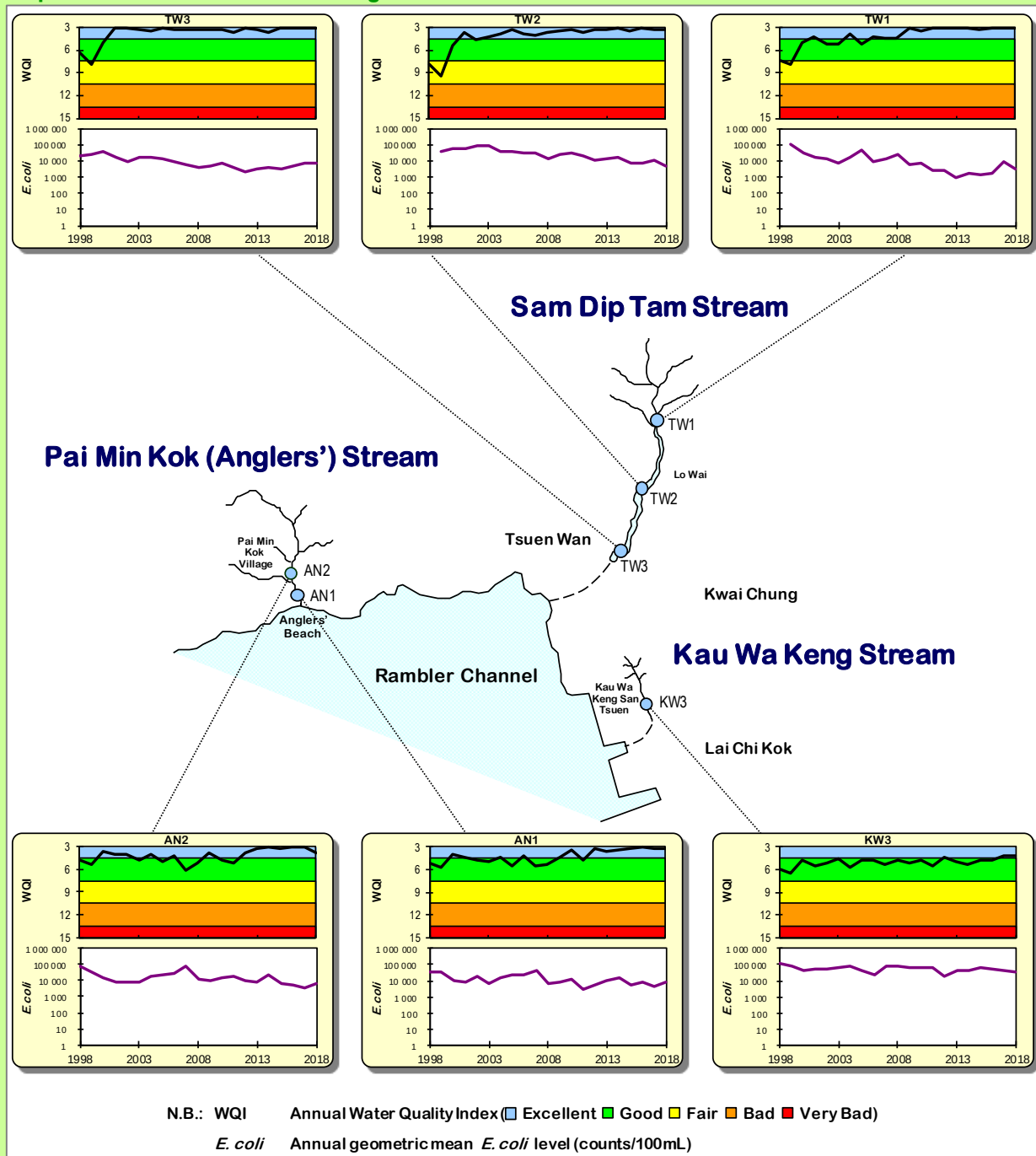
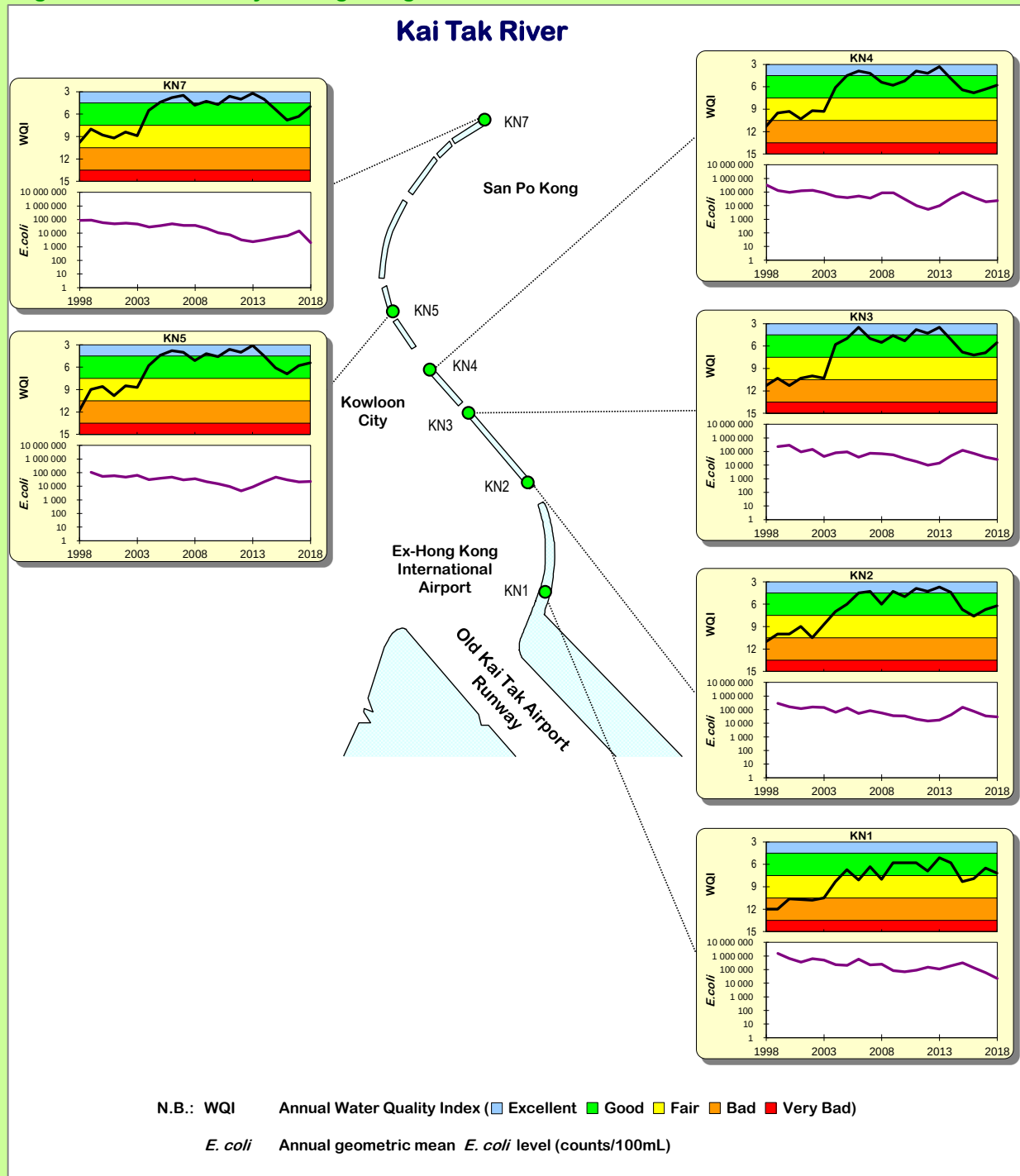


Figure 25. Water Quality Index gradings and *E. coli* levels in Kai Tak River



Appendices

Summary of river water quality monitoring stations and sampling frequencies in 2018

Area	Watercourse	Monitoring Station (Number)	Sampling frequency
Eastern New Territories			
Sha Tin	Shing Mun River	TR19I (1)	monthly
	<i>Shing Mun Main Channel</i>	TR23A, TR23L (2)	monthly
	<i>Siu Lek Yuen Nullah</i>	TR17, TR17L (2)	monthly
	<i>Fo Tan Nullah</i>	KY1 (1)	monthly
	<i>Kwun Yam Shan Stream</i>	TR19, TR19A, TR19C (3)	monthly
	<i>Tai Wai Nullah</i>	TR20B (1)	monthly
Tai Po Town Centre	Lam Tsuen River	TR12, TR12B, TR12C, TR12D, TR12E, TR12F, TR12G, TR12H, TR12I (9)	monthly
	Tai Po River	TR13 (1)	monthly
Tai Po Rural Area	Tai Po Kau Stream	TR14 (1)	monthly
	Shan Liu Stream	TR4 (1)	monthly
	Tung Tze Stream	TR6 (1)	monthly
Sai Kung	Ho Chung River	PR1, PR2 (2)	monthly
	Sha Kok Mei Stream	PR5, PR6 (2)	monthly
	Tai Chung Hau Stream	PR7, PR8 (2)	monthly
Tseung Kwan O	Tseng Lan Shue Stream	JR3, JR6, JR11 (3)	monthly
Northwestern New Territories			
North District	River Indus (Ng Tung River)	IN1, IN2, IN3 (3)	monthly
	River Beas (Sheung Yue River)	RB1, RB2, RB3 (3)	monthly
	River Ganges (Ping Yuen River)	GR1, GR2, GR3 (3)	monthly
Yuen Long	Yuen Long Creek	YL1, YL2, YL3, YL4 (4)	monthly
	Kam Tin River	KT1, KT2 (2)	monthly
	Tin Shui Wai Nullah	TSR1, TSR2 (2)	monthly
	Fairview Park Nullah	FVR1 (1)	monthly
Lau Fau Shan	Ha Pak Nai Stream	DB1 (1)	monthly
	Tai Shui Hang Stream	DB2 (1)	monthly
	Pak Nai Stream	DB3 (1)	monthly
	Sheung Pak Nai Stream	DB5 (1)	monthly
	Ngau Hom Sha Stream	DB6 (1)	monthly
	Tsang Kok Stream	DB8 (1)	monthly
Lantau Island			
Mui Wo	Mui Wo River	MW1, MW2, MW3, MW4, MW5 (5)	monthly
Tung Chung	Tung Chung River	TC1, TC2, TC3 (3)	monthly
Southwestern New Territories and Kowloon			
Tuen Mun	Tuen Mun River	TN1, TN2, TN3, TN4, TN5, TN6 (6)	monthly
Tsuen Wan and Kwai Chung	Pai Min Kok (Anglers') Stream	AN1, AN2 (2)	monthly
	Sam Dip Tam Stream	TW1, TW2, TW3 (3)	monthly
	Kau Wa Keng Stream	KW3 (1)	monthly
Kowloon	Kai Tak River	KN1, KN2, KN3, KN4, KN5, KN7 (6)	monthly
Total	30	82	-

River water quality parameters and the methods of analysis (part 1 of 2)

Water Quality Parameter	Reporting Limit and Unit	Analytical Method ¹ / Analyst
Physical Chemical Properties		
Water Temperature	0.1 °C	Multi-parameter water quality data logger, model YSI-6820 / On-site measurement / EPD
Dissolved Oxygen	0.1 mg/L, 1% saturation	
pH	0.1	
Conductivity	1 µS/cm	
Salinity	0.01, ppt	
Turbidity	0.1 NTU	
Flow	0.001 m³/s	Electromagnetic flow meter, model Flo-mate 2000 / Global Water Flow Probe, model FP211 / Electromagnetic flow meter, model Hach FH950 / On-site measurement / EPD
Solid Contents		
Suspended Solids	0.5 mg/L	In-house method GL-PH-23, based on APHA 22ed 2540 D & E / Government Laboratory
Total Solids	0.5 mg/L	In-house method GL-PH-19, based on APHA 20ed 2540 B & E / Government Laboratory
Total Volatile Solids	0.5 mg/L	In-house method GL-PH-19, based on APHA 20ed 2540 B & E / Government Laboratory
Aggregate Organics		
5-Day Biochemical Oxygen Demand (BOD ₅)	0.1 mg/L	In-house method based on APHA 18ed 5210 B / EPD
Chemical Oxygen Demand (COD)	2 mg/L	In-house method GL-OR-38, based on ASTM D1252-00, Method A or in-house method GL-OR-39, based on ASTM D1252-00, Method B / Government Laboratory
Total Organic Carbon (TOC)	1 mg/L	In-house method GL-OR-32, based on APHA 21ed 5310 B / Government Laboratory
Faecal Bacteria		
<i>Escherichia coli</i> (<i>E. coli</i>)	1 count/100 mL	In-house method ² , membrane filtration with CHROMagar Liquid ECC medium / EPD
Faecal Coliforms	1 count/100 mL	
Nutrients		
Ammonia-Nitrogen	0.005 mg/L	In-house method GL-IN-15, based on ASTM Standards, D 3590-11 Test Method B / Government Laboratory
Nitrite-Nitrogen	0.002 mg/L	In-house method GL-IN-18, based on APHA 22ed 4500-NO ₂ ⁻ B / Government Laboratory
Nitrate-Nitrogen	0.002 mg/L	In-house method GL-IN-18, based on APHA 22ed 4500-NO ₃ ⁻ I / Government Laboratory
Total Kjeldahl Nitrogen	0.05 mg/L	In-house method GL-IN-14 & GL-IN-15, based on ASTM D3590-11 Test Method B / Government Laboratory
Orthophosphate Phosphorus	0.002 mg/L	In-house method GL-IN-16, based on APHA 22ed 4500-P G / Government Laboratory
Total Phosphorus	0.02 mg/L	In-house method GL-IN-14 & GL-IN-16, based on APHA 22ed 4500-P G, ASTM D515-88 B (FIA) / Government Laboratory
Silica (as SiO ₂)	0.05 mg/L	In-house method GL-IN-17, based on APHA 22ed 4500-SiO ₂ F / Government Laboratory

Reference notes:

1. Mention of brand names of commercial products does not constitute or imply endorsement or recommendation by the Environmental Protection Department.
2. i) Ho, B.S.W. and Tam, T.Y. (1997). Enumeration of *E. coli* in environmental waters and wastewater using a chromogenic medium. *Wat. Sci. Tech.*, **35**, 409-413.
ii) DoE and DHSS (1983). "The bacteriological examination of drinking water supplies 1982. Report on Public Health and Medical Subjects No. 71. Methods for the Examination of Waters and Associated Materials". Department of Environment, Department of Health and Social Security, Public Health Laboratory Service, H.M.S.O. London.

River water quality parameters and the methods of analysis (part 2 of 2)

Water Quality Parameter	Reporting Limit and Unit	Analytical Method ¹ / Analyst
Metals		
Aluminium	50 µg/L	In-house method GL-TE-63, based on USEPA method 6020B (ICP-MS) / Government Laboratory
Antimony	1 µg/L	
Arsenic	1 µg/L	
Barium	1 µg/L	
Beryllium	1 µg/L	
Boron	50 µg/L	
Cadmium	0.1 µg/L	
Chromium	1 µg/L	
Copper	1 µg/L	
Iron	50 µg/L	
Lead	1 µg/L	
Manganese	10 µg/L	
Mercury	1 µg/L	
Molybdenum	2 µg/L	
Nickel	1 µg/L	
Silver	1 µg/L	
Thallium	1 µg/L	
Vanadium	2 µg/L	
Zinc	10 µg/L	
Industrial and Commercial Pollutants		
Cyanide	0.01 mg/L	In-house method GL-IN-42, based on ASTM D 4374-06 / Government Laboratory
Chloride	10 mg/L	In-house method GL-IN-43, based on APHA 20ed 4500-Cl ⁻ E & G / Government Laboratory
Fluoride	0.2 mg/L	In-house method GL-IN-47, based on APHA 20ed 4500-F ⁻ C & G / Government Laboratory
Anionic Surfactants (as Manoxol OT)	0.05 mg/L	In-house method GL-OR-30, based on BS 6068, Section 2.23 (1986), BS EN 903:1994, BS 6068: Section 2.23:1994 (Colorimetric) & In-house method GL-OR-65, based on Abbott, D.C. “Analyst”, Vol.87, p.286(1962) & S. Motomizu et al., “Analyst” Vol.113, p.747(1988) (FIA) / Government Laboratory
Oil and Grease	0.5 mg/L	In-house method GL-OR-26, based on APHA 20ed 5520 C / Government Laboratory
Sulphide Contents		
Free Hydrogen Sulphide	0.01 mg/L	In-house method GL-IN-46, based on APHA 20ed 4500S ²⁻ D / Government Laboratory
Sulphide	0.02 mg/L	
Plant Pigments		
Chlorophyll-a	0.2 µg/L	In house method GL-OR-34, based on APHA 20ed 10200H 2 / Government Laboratory
Phaeopigment	0.2 µg/L	

Key Water Quality Objectives (WQOs) for river monitoring stations in the Eastern New Territories

Watercourse	Monitoring station	Key Water Quality Objective				
		pH range	Maximum 5-Day Biochemical Oxygen Demand (mg/L)	Maximum Chemical Oxygen Demand (mg/L)	Maximum Annual Median Suspended Solids* (mg/L)	Minimum Dissolved Oxygen (mg/L)
Tolo Harbour and Channel Water Control Zone						
Shing Mun River	KY1	6.5 - 8.5	3	15	20	4
	TR17	6.5 - 8.5	5	30	20	4
	TR17L	6.5 - 8.5	5	30	20	4
	TR19	6.5 - 8.5	5	30	20	4
	TR19A	6.5 - 8.5	5	30	20	4
	TR19C	6.5 - 8.5	5	30	20	4
	TR19I	6.0 - 9.0	5	30	25	4
	TR20B	6.5 - 8.5	5	30	20	4
	TR23A	6.5 - 8.5	3	15	20	4
	TR23L	6.5 - 8.5	3	15	20	4
Lam Tsuen River	TR12	6.5 - 8.5	3	15	20	4
	TR12B	6.5 - 8.5	3	15	20	4
	TR12C	6.5 - 8.5	3	15	20	4
	TR12D	6.5 - 8.5	3	15	20	4
	TR12E	6.5 - 8.5	3	15	20	4
	TR12F	6.5 - 8.5	3	15	20	4
	TR12G	6.5 - 8.5	3	15	20	4
	TR12H	6.5 - 8.5	3	15	20	4
	TR12I	6.0 - 9.0	5	30	25	4
Tai Po River	TR13	6.5 - 8.5	5	30	20	4
Tai Po Kau Stream	TR14	6.0 - 9.0	5	30	25	4
Shan Liu Stream	TR4	6.0 - 9.0	5	30	25	4
Tung Tze Stream	TR6	6.0 - 9.0	5	30	25	4
Port Shelter Water Control Zone						
Ho Chung River	PR1	6.5 - 8.5	5	30	25	4
	PR2	6.5 - 8.5	5	30	25	4
Sha Kok Mei Stream	PR5	6.0 - 9.0	5	30	25	4
	PR6	6.0 - 9.0	5	30	25	4
Tai Chung Hau Stream	PR7	6.0 - 9.0	5	30	25	4
	PR8	6.0 - 9.0	5	30	25	4
Junk Bay Water Control Zone						
Tseng Lan Shue Stream	JR3	6.0 - 9.0	5	30	25	4
	JR6	6.0 - 9.0	5	30	25	4
	JR11	6.0 - 9.0	5	30	25	4

* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Key Water Quality Objectives (WQOs) for river monitoring stations in the Northwestern New Territories

Watercourse	Monitoring station	Key Water Quality Objective				
		pH range	Maximum 5-Day Biochemical Oxygen Demand (mg/L)	Maximum Chemical Oxygen Demand (mg/L)	Maximum Annual Median Suspended Solids* (mg/L)	Minimum Dissolved Oxygen (mg/L)
Deep Bay Water Control Zone						
River Indus	IN1	6.5 - 8.5	3	15	20	4
	IN2	6.5 - 8.5	3	15	20	4
	IN3	6.5 - 8.5	3	15	20	4
River Beas	RB1	6.5 - 8.5	3	15	20	4
	RB2	6.5 - 8.5	3	15	20	4
	RB3	6.5 - 8.5	3	15	20	4
River Ganges	GR1	6.5 - 8.5	3	15	20	4
	GR2	6.5 - 8.5	3	15	20	4
	GR3	6.5 - 8.5	3	15	20	4
Yuen Long Creek	YL1	6.5 - 8.5	3	15	20	4
	YL2	6.5 - 8.5	3	15	20	4
	YL3	6.5 - 8.5	5	30	20	4
	YL4	6.5 - 8.5	5	30	20	4
Kam Tin River	KT1	6.5 - 8.5	3	15	20	4
	KT2	6.5 - 8.5	3	15	20	4
Tin Shui Wai Nullah	TSR1	6.0 - 9.0	5	30	20	4
	TSR2	6.0 - 9.0	5	30	20	4
Fairview Park Nullah	FVR1	6.0 - 9.0	5	30	20	4
Ha Pak Nai Stream	DB1	6.0 - 9.0	5	30	20	4
Tai Shui Hang Stream	DB2	6.0 - 9.0	5	30	20	4
Pak Nai Stream	DB3	6.0 - 9.0	5	30	20	4
Sheung Pak Nai Stream	DB5	6.0 - 9.0	5	30	20	4
Ngau Hom Sha Stream	DB6	6.0 - 9.0	5	30	20	4
Tsang Kok Stream	DB8	6.0 - 9.0	5	30	20	4

* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Key Water Quality Objectives (WQOs) for river monitoring stations on Lantau Island

Watercourse	Monitoring station	Key Water Quality Objective				
		pH range	Maximum 5-Day Biochemical Oxygen Demand (mg/L)	Maximum Chemical Oxygen Demand (mg/L)	Maximum Annual Median Suspended Solids* (mg/L)	Minimum Dissolved Oxygen (mg/L)
Southern Water Control Zone						
Mui Wo River	MW1	6.5 - 8.5	5	30	20	4
	MW2	6.5 - 8.5	5	30	20	4
	MW3	6.5 - 8.5	5	30	20	4
	MW4	6.5 - 8.5	5	30	20	4
	MW5	6.0 - 9.0	5	30	25	4
North Western Water Control Zone						
Tung Chung River	TC1	6.0 - 9.0	5	30	25	4
	TC2	6.0 - 9.0	5	30	25	4
	TC3	6.0 - 9.0	5	30	25	4

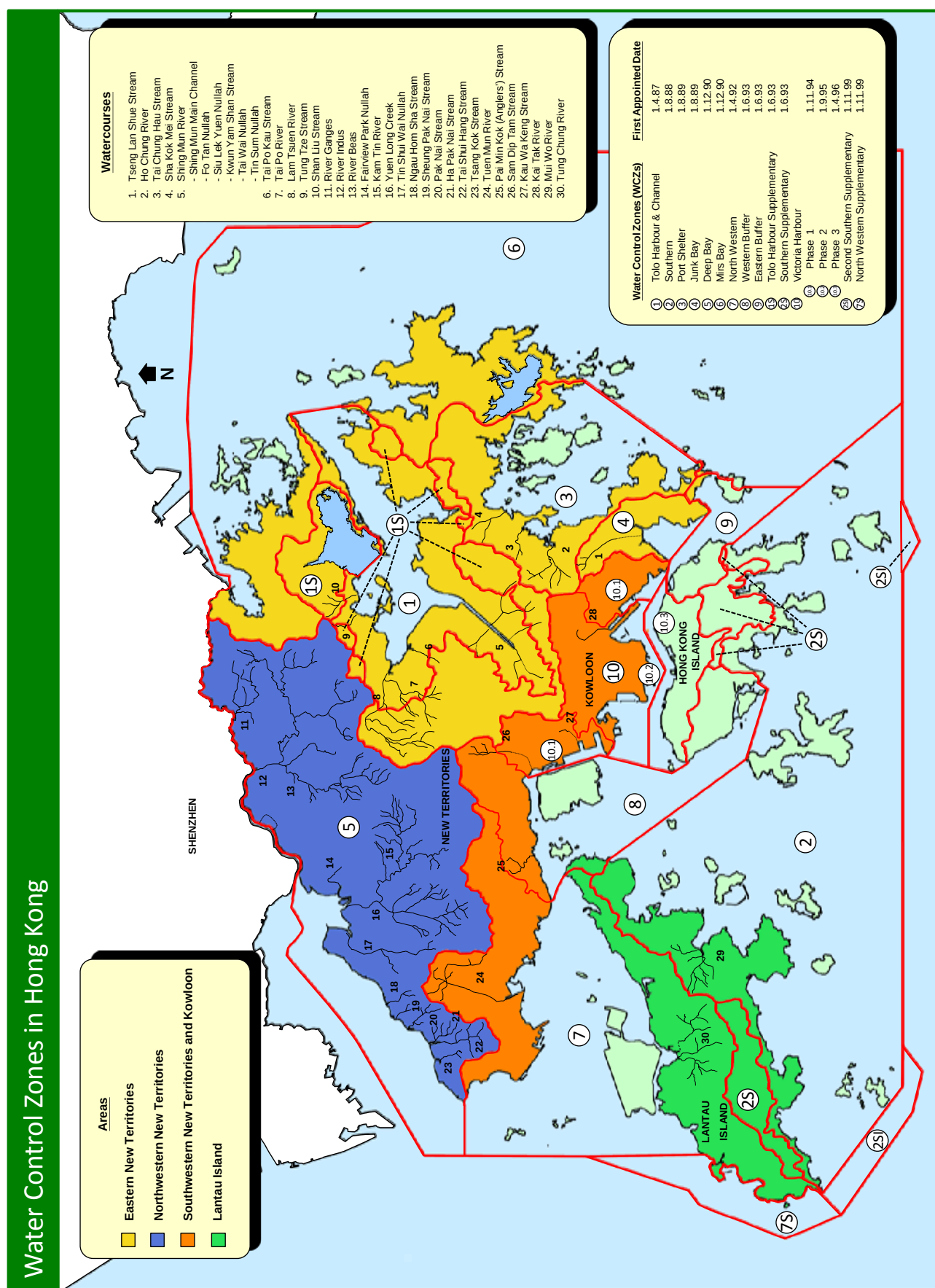
* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Key Water Quality Objectives (WQOs) for river monitoring stations in the Southwestern New Territories and Kowloon

Watercourse	Monitoring station	Key Water Quality Objective				
		pH range	Maximum 5-Day Biochemical Oxygen Demand (mg/L)	Maximum Chemical Oxygen Demand (mg/L)	Maximum Annual Median Suspended Solids (mg/L)	Minimum Dissolved Oxygen (mg/L)
North Western Water Control Zone						
Tuen Mun River	TN1	6.0 - 9.0	5	30	25	4
	TN2	6.5 - 8.5	3	15	20	4
	TN3	6.0 - 9.0	5	30	25	4
	TN4	6.0 - 9.0	5	30	25	4
	TN5	6.0 - 9.0	5	30	25	4
	TN6	6.0 - 9.0	5	30	25	4
Western Buffer Water Control Zone						
Pai Min Kok (Anglers') Stream	AN1	6.0 - 9.0	5	30	25	4
	AN2	6.0 - 9.0	5	30	25	4
Victoria Harbour Water Control Zone						
Sam Dip Tam Stream	TW1	6.0 - 9.0	5	30	25	4
	TW2	6.0 - 9.0	5	30	25	4
	TW3	6.0 - 9.0	5	30	25	4
Kau Wa Keng Stream	KW3	6.0 - 9.0	5	30	25	4
Kai Tak River	KN1	6.0 - 9.0	5	30	25	4
	KN2	6.0 - 9.0	5	30	25	4
	KN3	6.0 - 9.0	5	30	25	4
	KN4	6.0 - 9.0	5	30	25	4
	KN5	6.0 - 9.0	5	30	25	4
	KN7	6.0 - 9.0	5	30	25	4

* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Water Control Zones in Hong Kong



Summary of water quality monitoring data for Shing Mun River (Shing Mun Main Channel and Siu Lek Yuen Nullah) in 2018

Parameter	Unit	Shing Mun Main Channel	Siu Lek Yuen Nullah	
		TR19I	TR23L	TR23A
Dissolved Oxygen	mg/L	7.1 (5.9 - 8.8)	8.6 (7.7 - 10.3)	7.3 (5.2 - 9.8)
pH		8.0 (7.6 - 8.3)	8.4 (8.1 - 8.9)	7.9 (7.3 - 8.2)
Suspended Solids	mg/L	4.9 (2.6 - 16.0)	1.5 (0.6 - 2.8)	3.7 (1.4 - 6.9)
5-Day Biochemical Oxygen Demand	mg/L	3.1 (1.7 - 6.5)	0.5 (<0.1 - 1.5)	1.5 (0.6 - 3.9)
Chemical Oxygen Demand	mg/L	11 (6 - 34)	5 (<2 - 10)	11 (6 - 17)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	730 (50 - 36 000)	450 (10 - 90 000)	1 700 (120 - 83 000)
Faecal Coliforms	counts/ 100mL	8 600 (750 - 290 000)	6 000 (450 - 110 000)	12 000 (390 - 240 000)
Ammonia-Nitrogen	mg/L	0.115 (0.055 - 0.510)	0.021 (0.009 - 0.100)	0.145 (0.076 - 0.840)
Nitrate-Nitrogen	mg/L	0.175 (0.047 - 0.810)	0.300 (0.140 - 0.570)	0.435 (0.220 - 0.630)
Total Kjeldahl Nitrogen	mg/L	0.49 (0.32 - 1.30)	0.16 (<0.05 - 0.56)	0.37 (0.23 - 1.10)
Orthophosphate Phosphorus	mg/L	0.022 (<0.002 - 0.030)	0.004 (<0.002 - 0.024)	0.017 (0.003 - 0.061)
Total Phosphorus	mg/L	0.04 (0.03 - 0.07)	<0.02 (<0.02 - 0.03)	0.04 (0.02 - 0.10)
Sulphide	mg/L	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.02)
Aluminium	µg/L	<50 (<50 - 71)	<50 (<50 - 259)	<50 (<50 - 106)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	3 (1 - 5)	<1 (<1 - <1)	2 (<1 - 3)
Copper	µg/L	5 (4 - 15)	1 (<1 - 5)	2 (2 - 5)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	<10 (<10 - 28)	<10 (<10 - 25)	<10 (<10 - 37)
Flow	m ³ /s	NM	0.034 (0.015 - 0.090)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Shing Mun River (Fo Tan Nullah and Kwun Yam Shan Stream) in 2018

Parameter	Unit	<u>Fo Tan Nullah</u>		<u>Kwun Yam Shan Stream</u>
		TR17	TR17L	KY1
Dissolved Oxygen	mg/L	8.2 (6.7 - 11.6)	6.7 (4.3 - 9.6)	8.3 (7.5 - 10.4)
pH		8.8 (7.4 - 10.7)	7.9 (7.4 - 9.2)	7.2 (6.8 - 8.4)
Suspended Solids	mg/L	7.0 (2.1 - 490.0)	7.6 (2.2 - 83.0)	4.1 (1.0 - 16.0)
5-Day Biochemical Oxygen Demand	mg/L	6.2 (1.4 - 110.0)	3.0 (1.7 - 18.0)	0.4 (<0.1 - 1.7)
Chemical Oxygen Demand	mg/L	19 (4 - 390)	13 (8 - 37)	5 (<2 - 16)
Oil & Grease	mg/L	<0.5 (<0.5 - 16.0)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	19 000 (600 - 250 000)	4 500 (580 - 38 000)	290 (22 - 3 900)
Faecal Coliforms	counts/ 100mL	130 000 (2 400 - 2 500 000)	39 000 (8 300 - 250 000)	1 100 (35 - 20 000)
Ammonia-Nitrogen	mg/L	0.270 (0.018 - 2.900)	0.375 (0.160 - 0.990)	0.019 (0.011 - 0.075)
Nitrate-Nitrogen	mg/L	0.510 (<0.002 - 1.400)	0.490 (0.190 - 1.000)	0.550 (0.240 - 0.860)
Total Kjeldahl Nitrogen	mg/L	1.40 (0.25 - 6.50)	0.63 (0.38 - 1.70)	0.25 (0.11 - 0.43)
Orthophosphate Phosphorus	mg/L	0.014 (<0.002 - 0.040)	0.019 (0.007 - 0.080)	0.081 (0.044 - 0.110)
Total Phosphorus	mg/L	0.09 (0.02 - 0.81)	0.06 (0.03 - 0.17)	0.10 (0.06 - 0.16)
Sulphide	mg/L	<0.02 (<0.02 - 0.15)	<0.02 (<0.02 - 0.15)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	99 (<50 - 1 013)	<50 (<50 - 100)	108 (<50 - 235)
Cadmium	µg/L	<0.1 (<0.1 - 0.2)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	2 (<1 - 6)	3 (1 - 6)	<1 (<1 - <1)
Copper	µg/L	3 (<1 - 19)	3 (2 - 8)	<1 (<1 - 2)
Lead	µg/L	<1 (<1 - 5)	<1 (<1 - 1)	<1 (<1 - <1)
Zinc	µg/L	13 (<10 - 89)	12 (<10 - 34)	<10 (<10 - <10)
Flow	m ³ /s	0.163 (0.025 - 1.250)	NM	0.016 (0.000 - 0.150)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Shing Mun River (Tai Wai Nullah and Tin Sum Nullah) in 2018

Parameter	Unit	Tai Wai Nullah		Tin Sum Nullah	
		TR19A	TR19C	TR19	TR20B
Dissolved Oxygen	mg/L	8.8 (8.0 - 10.4)	8.9 (7.7 - 10.9)	9.7 (8.2 - 11.5)	8.4 (7.6 - 9.6)
pH		7.7 (7.2 - 8.9)	7.1 (6.9 - 7.9)	7.3 (7.1 - 8.6)	7.0 (6.8 - 7.8)
Suspended Solids	mg/L	3.5 (1.3 - 15.0)	2.3 (1.1 - 7.0)	2.3 (1.0 - 6.0)	2.6 (1.0 - 11.0)
5-Day Biochemical Oxygen Demand	mg/L	1.2 (<0.1 - 7.4)	1.2 (0.6 - 5.0)	1.7 (0.6 - 5.0)	<0.1 (<0.1 - 0.4)
Chemical Oxygen Demand	mg/L	7 (4 - 31)	7 (3 - 21)	7 (3 - 17)	5 (<2 - 10)
Oil & Grease	mg/L	<0.5 (<0.5 - 0.8)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	1 800 (<1 - 130 000)	5 800 (310 - 90 000)	3 700 (170 - 35 000)	<1 (<1 - <1)
Faecal Coliforms	counts/ 100mL	9 400 (<1 - 330 000)	31 000 (7 100 - 460 000)	43 000 (8 500 - 210 000)	1 (<1 - 9)
Ammonia-Nitrogen	mg/L	0.068 (0.009 - 0.180)	0.067 (0.042 - 0.220)	0.069 (0.020 - 0.230)	0.041 (0.014 - 0.720)
Nitrate-Nitrogen	mg/L	0.965 (0.740 - 1.200)	0.915 (0.690 - 1.000)	0.905 (0.690 - 1.000)	1.300 (0.690 - 1.900)
Total Kjeldahl Nitrogen	mg/L	0.39 (0.22 - 1.50)	0.34 (0.22 - 1.10)	0.38 (0.17 - 1.10)	0.26 (0.10 - 1.20)
Orthophosphate Phosphorus	mg/L	0.021 (0.007 - 0.035)	0.020 (0.014 - 0.032)	0.016 (0.004 - 0.042)	0.008 (<0.002 - 0.064)
Total Phosphorus	mg/L	0.04 (<0.02 - 0.16)	0.03 (<0.02 - 0.07)	0.04 (0.02 - 0.08)	<0.02 (<0.02 - 0.09)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	53 (<50 - 214)	<50 (<50 - 190)	<50 (<50 - 211)	64 (<50 - 763)
Cadmium	µg/L	0.2 (<0.1 - 0.4)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - 1)	<1 (<1 - 2)	<1 (<1 - 2)
Copper	µg/L	2 (<1 - 14)	2 (<1 - 15)	3 (<1 - 13)	2 (<1 - 3)
Lead	µg/L	<1 (<1 - 3)	<1 (<1 - 3)	<1 (<1 - 2)	<1 (<1 - 7)
Zinc	µg/L	16 (<10 - 94)	16 (<10 - 78)	11 (<10 - 85)	<10 (<10 - 49)
Flow	m ³ /s	0.034 (0.015 - 0.068)	0.090 (0.045 - 0.330)	0.099 (0.030 - 0.900)	0.020 (0.012 - 0.090)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Lam Tsuen River in 2018 (Part 1 of 3)

Parameter	Unit	<u>Lam Tsuen River</u>		
		TR12H	TR12D	TR12C
Dissolved Oxygen	mg/L	8.6 (7.8 - 10.8)	8.7 (8.0 - 10.8)	8.3 (5.3 - 10.1)
pH		7.0 (6.3 - 7.5)	7.1 (6.6 - 7.6)	7.2 (7.0 - 7.5)
Suspended Solids	mg/L	1.4 (0.5 - 4.7)	0.9 (<0.5 - 5.2)	2.9 (0.6 - 9.2)
5-Day Biochemical Oxygen Demand	mg/L	0.4 (<0.1 - 0.8)	0.2 (<0.1 - 0.6)	1.3 (0.6 - 3.0)
Chemical Oxygen Demand	mg/L	2 (<2 - 6)	3 (<2 - 6)	5 (4 - 17)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	1 100 (410 - 6 100)	110 (7 - 640)	4 500 (1 300 - 34 000)
Faecal Coliforms	counts/ 100mL	4 800 (1 100 - 24 000)	1 200 (320 - 4 600)	18 000 (4 900 - 68 000)
Ammonia-Nitrogen	mg/L	0.125 (0.034 - 0.280)	0.016 (0.012 - 0.026)	0.140 (0.062 - 0.350)
Nitrate-Nitrogen	mg/L	0.730 (0.550 - 0.950)	0.360 (0.092 - 0.710)	1.015 (0.500 - 1.800)
Total Kjeldahl Nitrogen	mg/L	0.33 (0.11 - 0.46)	0.14 (<0.05 - 0.24)	0.46 (0.26 - 0.75)
Orthophosphate Phosphorus	mg/L	0.037 (0.005 - 0.059)	0.008 (<0.002 - 0.014)	0.061 (0.012 - 0.086)
Total Phosphorus	mg/L	0.05 (0.03 - 0.07)	<0.02 (<0.02 - 0.03)	0.11 (0.06 - 0.14)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - <50)	<50 (<50 - <50)	<50 (<50 - 67)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	1 (<1 - 3)	<1 (<1 - 4)	1 (<1 - 2)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - 1)
Zinc	µg/L	<10 (<10 - 31)	10 (<10 - 47)	<10 (<10 - 23)
Flow	m ³ /s	0.110 (0.040 - 0.405)	0.030 (0.010 - 0.100)	0.120 (0.060 - 0.210)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Lam Tsuen River in 2018 (Part 2 of 3)

Parameter	Unit	<u>Lam Tsuen River</u>		
		TR12G	TR12F	TR12B
Dissolved Oxygen	mg/L	8.4 (6.9 - 10.4)	8.7 (7.6 - 10.8)	8.9 (8.0 - 11.2)
pH		6.7 (6.3 - 7.7)	7.1 (6.4 - 7.7)	7.3 (6.9 - 7.6)
Suspended Solids	mg/L	1.7 (0.7 - 3.4)	1.4 (0.5 - 8.0)	1.9 (0.5 - 4.8)
5-Day Biochemical Oxygen Demand	mg/L	0.3 (<0.1 - 0.4)	0.3 (<0.1 - 0.5)	0.5 (0.3 - 2.1)
Chemical Oxygen Demand	mg/L	4 (<2 - 16)	6 (<2 - 14)	4 (<2 - 11)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	180 (68 - 570)	260 (84 - 1 100)	390 (150 - 1 900)
Faecal Coliforms	counts/ 100mL	1 500 (130 - 8 800)	3 000 (590 - 11 000)	2 800 (780 - 9 100)
Ammonia-Nitrogen	mg/L	0.017 (<0.005 - 0.026)	0.025 (0.008 - 0.042)	0.039 (0.026 - 0.058)
Nitrate-Nitrogen	mg/L	0.059 (<0.002 - 0.120)	0.280 (0.180 - 0.490)	0.850 (0.280 - 1.400)
Total Kjeldahl Nitrogen	mg/L	0.16 (0.06 - 0.30)	0.23 (0.06 - 0.34)	0.23 (0.14 - 0.44)
Orthophosphate Phosphorus	mg/L	0.014 (<0.002 - 0.029)	0.015 (<0.002 - 0.049)	0.030 (0.005 - 0.068)
Total Phosphorus	mg/L	0.03 (0.02 - 0.06)	0.04 (0.02 - 0.08)	0.04 (0.03 - 0.08)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 63)	<50 (<50 - 113)	<50 (<50 - 63)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	<1 (<1 - <1)	<1 (<1 - 1)	1 (<1 - 3)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	<10 (<10 - 13)	<10 (<10 - 11)	<10 (<10 - 13)
Flow	m ³ /s	0.020 (0.010 - 0.060)	0.055 (0.020 - 0.135)	0.420 (0.168 - 2.100)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Lam Tsuen River (Part 3 of 3) and Tai Po River in 2018

Parameter	Unit	Lam Tsuen River			Tai Po River
		TR12E	TR12	TR12I	TR13
Dissolved Oxygen	mg/L	8.5 (7.8 - 10.2)	8.4 (6.4 - 10.7)	5.6 (4.5 - 8.1)	8.6 (7.4 - 10.9)
pH		8.0 (7.6 - 8.1)	7.3 (7.1 - 8.1)	7.2 (7.1 - 7.8)	7.1 (6.5 - 8.2)
Suspended Solids	mg/L	1.7 (0.7 - 6.0)	3.7 (0.6 - 20.0)	3.3 (2.0 - 7.2)	2.9 (<0.5 - 29.0)
5-Day Biochemical Oxygen Demand	mg/L	0.3 (<0.1 - 0.7)	3.7 (1.9 - 11.0)	1.7 (0.2 - 3.5)	0.9 (0.2 - 3.4)
Chemical Oxygen Demand	mg/L	5 (<2 - 7)	14 (6 - 31)	11 (7 - 13)	6 (2 - 18)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	660 (16 - 9 800)	2 400 (360 - 20 000)	36 000 (6 500 - 190 000)	9 800 (2 800 - 52 000)
Faecal Coliforms	counts/ 100mL	3 000 (490 - 14 000)	11 000 (1 400 - 37 000)	110 000 (23 000 - 450 000)	28 000 (8 100 - 160 000)
Ammonia-Nitrogen	mg/L	0.026 (<0.005 - 0.095)	0.565 (0.200 - 4.500)	0.515 (0.310 - 0.800)	0.069 (0.025 - 0.260)
Nitrate-Nitrogen	mg/L	0.905 (0.480 - 1.200)	1.700 (0.790 - 3.200)	0.790 (0.540 - 0.990)	0.680 (0.290 - 1.900)
Total Kjeldahl Nitrogen	mg/L	0.21 (0.09 - 0.37)	1.55 (0.34 - 4.90)	0.79 (0.50 - 1.30)	0.37 (0.16 - 1.10)
Orthophosphate Phosphorus	mg/L	0.009 (<0.002 - 0.036)	0.320 (0.007 - 0.700)	0.072 (0.008 - 0.120)	0.031 (0.005 - 0.098)
Total Phosphorus	mg/L	0.02 (<0.02 - 0.050)	0.45 (0.09 - 0.95)	0.11 (0.07 - 0.17)	0.05 (0.02 - 0.16)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	88 (<50 - 284)	<50 (<50 - 59)	<50 (<50 - 95)	<50 (<50 - 474)
Cadmium	µg/L	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - 1)	2 (<1 - 4)	<1 (<1 - 5)
Copper	µg/L	<1 (<1 - 2)	2 (<1 - 4)	2 (2 - 5)	1 (<1 - 4)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - 1)
Zinc	µg/L	<10 (<10 - 26)	<10 (<10 - 19)	<10 (<10 - 19)	<10 (<10 - 18)
Flow	m ³ /s	0.240 (0.165 - 0.325)	0.123 (0.038 - 0.675)	NM	0.263 (0.075 - 2.250)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tai Po Kau Stream, Shan Liu Stream and Tung Tze Stream in 2018

Parameter	Unit	<u>Tai Po Kau Stream</u>	<u>Shan Liu Stream</u>	<u>Tung Tze Stream</u>
		TR14	TR4	TR6
Dissolved Oxygen	mg/L	8.2 (7.0 - 10.3)	8.7 (7.4 - 10.2)	6.0 (4.0 - 8.2)
pH		7.0 (6.6 - 7.4)	7.4 (6.7 - 8.2)	7.5 (7.0 - 8.1)
Suspended Solids	mg/L	1.8 (0.5 - 22.0)	2.9 (0.8 - 38.0)	9.3 (1.3 - 32.0)
5-Day Biochemical Oxygen Demand	mg/L	0.3 (<0.1 - 1.5)	0.7 (0.4 - 3.5)	1.5 (0.6 - 3.3)
Chemical Oxygen Demand	mg/L	6 (<2 - 12)	6 (<2 - 11)	15 (9 - 34)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	1 200 (67 - 34 000)	2 900 (360 - 90 000)	4 500 (200 - 80 000)
Faecal Coliforms	counts/ 100mL	3 700 (790 - 64 000)	10 000 (2 000 - 240 000)	8 300 (240 - 210 000)
Ammonia-Nitrogen	mg/L	0.075 (0.029 - 0.250)	0.081 (0.031 - 0.260)	0.515 (0.057 - 3.000)
Nitrate-Nitrogen	mg/L	0.275 (0.130 - 1.200)	0.910 (0.420 - 1.600)	0.400 (<0.002 - 0.790)
Total Kjeldahl Nitrogen	mg/L	0.20 (0.10 - 0.69)	0.30 (0.19 - 1.00)	0.93 (0.25 - 3.30)
Orthophosphate Phosphorus	mg/L	0.012 (0.004 - 0.018)	0.048 (0.021 - 0.120)	0.051 (0.011 - 0.230)
Total Phosphorus	mg/L	<0.02 (<0.02 - 0.08)	0.06 (0.04 - 0.18)	0.10 (0.04 - 0.31)
Sulphide	mg/L	<0.02 (<0.02 - 0.04)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.04)
Aluminium	µg/L	<50 (<50 - 538)	<50 (<50 - 209)	<50 (<50 - 260)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - <1)	2 (<1 - 4)
Copper	µg/L	<1 (<1 - 2)	<1 (<1 - 3)	3 (1 - 5)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - 1)	<1 (<1 - 1)
Zinc	µg/L	<10 (<10 - 20)	<10 (<10 - 17)	<10 (<10 - 19)
Flow	m ³ /s	0.070 (0.025 - 0.480)	0.060 (0.010 - 0.125)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Ho Chung River in 2018

Parameter	Unit	<u>Ho Chung River</u>	
		PR1	PR2
Dissolved Oxygen	mg/L	7.4 (5.9 - 7.8)	8.4 (7.2 - 8.9)
pH		7.4 (7.0 - 7.5)	7.2 (6.8 - 7.4)
Suspended Solids	mg/L	3.5 (2.1 - 13.0)	3.3 (2.1 - 5.4)
5-Day Biochemical Oxygen Demand	mg/L	1.3 (0.6 - 4.3)	0.6 (0.3 - 2.7)
Chemical Oxygen Demand	mg/L	10 (5 - 22)	3 (<2 - 8)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	4 300 (1 400 - 40 000)	1 100 (380 - 3 200)
Faecal Coliforms	counts/ 100mL	12 000 (3 100 - 68 000)	6 000 (2 900 - 15 000)
Ammonia-Nitrogen	mg/L	0.495 (0.088 - 2.200)	0.081 (0.020 - 0.260)
Nitrate-Nitrogen	mg/L	0.335 (0.170 - 0.570)	0.265 (0.140 - 0.410)
Total Kjeldahl Nitrogen	mg/L	1.07 (0.34 - 2.40)	0.26 (0.15 - 0.56)
Orthophosphate Phosphorus	mg/L	0.038 (0.007 - 0.150)	0.012 (0.005 - 0.044)
Total Phosphorus	mg/L	0.09 (0.03 - 0.47)	0.03 (<0.02 - 0.06)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 64)	53 (<50 - 93)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	2 (<1 - 4)	<1 (<1 - <1)
Copper	µg/L	2 (<1 - 5)	<1 (<1 - 2)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	<10 (<10 - 22)	<10 (<10 - 18)
Flow	m ³ /s	NM	0.300 (0.200 - 0.700)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Sha Kok Mei Stream in 2018

Parameter	Unit	<u>Sha Kok Mei Stream</u>	
		PR5	PR6
Dissolved Oxygen	mg/L	7.2 (5.3 - 9.4)	8.3 (7.4 - 9.3)
pH		7.4 (7.1 - 8.0)	7.1 (6.7 - 7.3)
Suspended Solids	mg/L	2.0 (0.9 - 11.0)	1.7 (1.0 - 17.0)
5-Day Biochemical Oxygen Demand	mg/L	1.5 (<0.1 - 6.9)	1.3 (0.2 - 4.0)
Chemical Oxygen Demand	mg/L	5 (3 - 16)	6 (3 - 11)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	200 (<10 - 23 000)	9 400 (2 900 - 39 000)
Faecal Coliforms	counts/ 100mL	600 (<10 - 57 000)	19 000 (4 600 - 53 000)
Ammonia-Nitrogen	mg/L	0.215 (0.012 - 0.850)	0.175 (0.058 - 0.510)
Nitrate-Nitrogen	mg/L	0.835 (0.470 - 1.400)	1.550 (1.200 - 2.300)
Total Kjeldahl Nitrogen	mg/L	0.72 (0.18 - 1.30)	0.50 (0.34 - 0.68)
Orthophosphate Phosphorus	mg/L	0.043 (0.004 - 0.160)	0.085 (0.030 - 0.150)
Total Phosphorus	mg/L	0.09 (<0.02 - 0.23)	0.11 (0.05 - 0.19)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 80)	<50 (<50 - 57)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	<1 (<1 - 2)	1 (<1 - 2)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	<10 (<10 - 27)	<10 (<10 - 17)
Flow	m ³ /s	0.095 (0.045 - 0.150)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tai Chung Hau Stream in 2018

Parameter	Unit	Tai Chung Hau Stream	
		PR7	PR8
Dissolved Oxygen	mg/L	8.5 (7.0 - 9.9)	8.2 (7.4 - 9.2)
pH		7.7 (7.3 - 8.4)	7.6 (7.2 - 9.0)
Suspended Solids	mg/L	2.2 (1.2 - 3.4)	2.3 (1.3 - 13.0)
5-Day Biochemical Oxygen Demand	mg/L	0.9 (0.4 - 2.5)	2.2 (1.1 - 6.1)
Chemical Oxygen Demand	mg/L	8 (4 - 17)	7 (3 - 10)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	3 500 (800 - 21 000)	14 000 (3 700 - 37 000)
Faecal Coliforms	counts/ 100mL	14 000 (1 200 - 50 000)	38 000 (15 000 - 72 000)
Ammonia-Nitrogen	mg/L	0.069 (0.037 - 0.310)	0.055 (0.006 - 0.300)
Nitrate-Nitrogen	mg/L	0.570 (0.310 - 1.000)	0.875 (0.460 - 1.200)
Total Kjeldahl Nitrogen	mg/L	0.34 (0.20 - 0.75)	0.38 (0.21 - 0.66)
Orthophosphate Phosphorus	mg/L	0.041 (0.008 - 0.058)	0.050 (0.007 - 0.080)
Total Phosphorus	mg/L	0.07 (0.04 - 0.09)	0.08 (0.04 - 0.14)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 172)	<50 (<50 - 119)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 4)	<1 (<1 - 3)
Copper	µg/L	1 (<1 - 5)	1 (<1 - 2)
Lead	µg/L	<1 (<1 - 1)	<1 (<1 - <1)
Zinc	µg/L	<10 (<10 - 29)	<10 (<10 - 23)
Flow	m ³ /s	0.160 (0.040 - 0.560)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tseng Lan Shue Stream in 2018

Parameter	Unit	Tseng Lan Shue Stream		
		JR3	JR6	JR11
Dissolved Oxygen	mg/L	5.7 (3.9 - 7.6)	7.7 (6.9 - 8.5)	8.9 (8.0 - 10.1)
pH		7.0 (6.6 - 8.8)	7.2 (6.9 - 7.5)	7.0 (6.4 - 7.5)
Suspended Solids	mg/L	6.0 (1.8 - 120.0)	5.1 (1.4 - 63.0)	2.5 (1.4 - 8.0)
5-Day Biochemical Oxygen Demand	mg/L	10.5 (3.9 - 60.0)	4.7 (1.6 - 23.0)	1.0 (0.6 - 4.2)
Chemical Oxygen Demand	mg/L	24 (7 - 41)	14 (5 - 26)	8 (4 - 10)
Oil & Grease	mg/L	0.7 (<0.5 - 36.0)	<0.5 (<0.5 - 0.6)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	100 000 (6 000 - 310 000)	31 000 (6 800 - 100 000)	1 000 (210 - 2 700)
Faecal Coliforms	counts/ 100mL	160 000 (15 000 - 490 000)	57 000 (10 000 - 220 000)	3 200 (750 - 8 800)
Ammonia-Nitrogen	mg/L	6.400 (1.900 - 13.000)	0.330 (0.041 - 0.700)	0.061 (0.025 - 0.170)
Nitrate-Nitrogen	mg/L	1.300 (<0.002 - 1.700)	2.500 (1.300 - 3.200)	4.300 (1.400 - 5.600)
Total Kjeldahl Nitrogen	mg/L	10.75 (2.40 - 14.00)	1.30 (0.50 - 3.70)	0.44 (<0.05 - 0.77)
Orthophosphate Phosphorus	mg/L	0.520 (0.180 - 1.000)	0.415 (0.180 - 0.630)	0.455 (0.130 - 0.760)
Total Phosphorus	mg/L	0.73 (0.25 - 1.20)	0.56 (0.23 - 1.10)	0.50 (0.15 - 0.92)
Sulphide	mg/L	<0.02 (<0.02 - 0.06)	<0.02 (<0.02 - 0.05)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 131)	<50 (<50 - 295)	<50 (<50 - 79)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.2)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 5)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	3 (1 - 5)	2 (<1 - 8)	1 (<1 - 3)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - 4)	<1 (<1 - <1)
Zinc	µg/L	14 (<10 - 32)	10 (<10 - 64)	<10 (<10 - 30)
Flow	m ³ /s	NM	NM	0.100 (0.080 - 0.160)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for River Indus in 2018

Parameter	Unit	River Indus		
		IN1	IN2	IN3
Dissolved Oxygen	mg/L	5.3 (3.4 - 6.6)	7.4 (5.5 - 12.2)	8.2 (6.0 - 8.9)
pH		6.9 (6.7 - 7.2)	7.0 (6.7 - 8.7)	8.1 (7.3 - 10.3)
Suspended Solids	mg/L	38.5 (8.0 - 85.0)	7.6 (4.2 - 29.0)	4.4 (1.8 - 390.0)
5-Day Biochemical Oxygen Demand	mg/L	4.3 (2.7 - 12.0)	2.3 (0.8 - 12.0)	1.0 (0.2 - 4.1)
Chemical Oxygen Demand	mg/L	23 (10 - 47)	12 (5 - 25)	9 (3 - 17)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	29 000 (6 600 - 230 000)	2 000 (170 - 6 600)	590 (30 - 23 000)
Faecal Coliforms	counts/ 100mL	77 000 (13 000 - 480 000)	8 200 (2 100 - 27 000)	2 700 (80 - 54 000)
Ammonia-Nitrogen	mg/L	1.250 (0.810 - 3.700)	0.825 (0.170 - 1.600)	0.094 (0.032 - 0.480)
Nitrate-Nitrogen	mg/L	3.550 (0.660 - 6.800)	1.200 (0.610 - 1.400)	0.535 (0.440 - 0.900)
Total Kjeldahl Nitrogen	mg/L	2.70 (1.70 - 6.00)	1.45 (0.63 - 2.70)	0.57 (0.28 - 2.50)
Orthophosphate Phosphorus	mg/L	0.180 (0.073 - 0.350)	0.049 (0.024 - 0.100)	0.057 (0.030 - 0.140)
Total Phosphorus	mg/L	0.35 (0.20 - 0.83)	0.15 (0.10 - 0.19)	0.12 (0.06 - 0.27)
Sulphide	mg/L	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	54 (<50 - 492)	<50 (<50 - 160)	355 (<50 - 2 779)
Cadmium	µg/L	<0.1 (<0.1 - 0.6)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 4)	<1 (<1 - <1)	2 (<1 - 31)
Copper	µg/L	2 (2 - 8)	1 (<1 - 2)	2 (1 - 6)
Lead	µg/L	<1 (<1 - 5)	<1 (<1 - 1)	<1 (<1 - 1)
Zinc	µg/L	14 (<10 - 65)	<10 (<10 - 20)	<10 (<10 - <10)
Flow	m ³ /s	NM	NM	0.060 (0.015 - 0.200)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for River Beas in 2018

Parameter	Unit	River Beas		
		RB1	RB2	RB3
Dissolved Oxygen	mg/L	9.3 (7.6 - 11.0)	8.2 (7.0 - 11.5)	7.7 (4.3 - 12.0)
pH		7.6 (6.7 - 9.0)	7.1 (6.4 - 7.6)	7.2 (6.8 - 7.6)
Suspended Solids	mg/L	6.9 (2.1 - 390.0)	5.6 (2.8 - 38.0)	17.0 (4.0 - 270.0)
5-Day Biochemical Oxygen Demand	mg/L	4.9 (1.2 - 11.0)	3.6 (0.9 - 14.0)	5.3 (1.0 - 24.0)
Chemical Oxygen Demand	mg/L	11 (4 - 23)	10 (5 - 34)	23 (7 - 52)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	4 400 (900 - 80 000)	2 300 (80 - 17 000)	12 000 (660 - 66 000)
Faecal Coliforms	counts/ 100mL	20 000 (5 300 - 860 000)	16 000 (630 - 240 000)	45 000 (3 700 - 120 000)
Ammonia-Nitrogen	mg/L	0.285 (0.098 - 1.200)	1.700 (0.160 - 3.000)	1.550 (0.230 - 4.700)
Nitrate-Nitrogen	mg/L	0.915 (0.430 - 1.600)	0.465 (0.270 - 0.810)	0.795 (0.140 - 6.000)
Total Kjeldahl Nitrogen	mg/L	0.92 (0.32 - 2.00)	2.20 (0.87 - 4.10)	3.00 (0.92 - 6.00)
Orthophosphate Phosphorus	mg/L	0.225 (0.039 - 0.430)	0.155 (0.032 - 0.280)	0.145 (0.034 - 1.000)
Total Phosphorus	mg/L	0.35 (0.10 - 0.71)	0.30 (0.10 - 0.63)	0.44 (0.11 - 1.20)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - 0.03)
Aluminium	µg/L	<50 (<50 - 349)	<50 (<50 - 153)	<50 (<50 - 148)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.2)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	1 (<1 - 2)	1 (<1 - 12)	2 (2 - 13)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - 1)	<1 (<1 - 4)
Zinc	µg/L	<10 (<10 - 18)	<10 (<10 - 16)	19 (<10 - 61)
Flow	m ³ /s	0.115 (0.040 - 1.500)	0.200 (0.034 - 0.546)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for River Ganges in 2018

Parameter	Unit	River Ganges		
		GR1	GR2	GR3
Dissolved Oxygen	mg/L	8.4 (6.8 - 14.0)	5.5 (3.1 - 7.4)	8.0 (5.6 - 9.5)
pH		7.4 (6.9 - 8.6)	6.8 (6.5 - 7.3)	7.0 (6.5 - 7.5)
Suspended Solids	mg/L	11.0 (5.4 - 69.0)	9.5 (3.8 - 29.0)	5.6 (<0.5 - 24.0)
5-Day Biochemical Oxygen Demand	mg/L	6.6 (1.6 - 26.0)	4.2 (1.5 - 7.5)	0.7 (0.1 - 2.6)
Chemical Oxygen Demand	mg/L	16 (11 - 34)	12 (7 - 21)	5 (2 - 13)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.6)	<0.5 (<0.5 - 0.6)
<i>E. coli</i>	counts/ 100mL	4 100 (700 - 70 000)	3 400 (380 - 16 000)	400 (80 - 26 000)
Faecal Coliforms	counts/ 100mL	11 000 (1 300 - 81 000)	13 000 (930 - 74 000)	7 400 (450 - 230 000)
Ammonia-Nitrogen	mg/L	5.500 (0.310 - 32.000)	5.300 (0.240 - 26.000)	0.077 (0.019 - 0.380)
Nitrate-Nitrogen	mg/L	1.150 (0.900 - 2.500)	0.865 (0.027 - 1.700)	0.265 (0.170 - 0.580)
Total Kjeldahl Nitrogen	mg/L	6.40 (1.00 - 36.00)	6.00 (0.40 - 28.00)	0.51 (<0.05 - 1.10)
Orthophosphate Phosphorus	mg/L	0.480 (0.160 - 3.000)	0.180 (0.067 - 0.730)	0.004 (<0.002 - 0.006)
Total Phosphorus	mg/L	0.66 (0.32 - 3.10)	0.36 (0.20 - 1.20)	<0.02 (<0.02 - 0.06)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	52 (<50 - 111)	51 (<50 - 538)	<50 (<50 - 119)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.4)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	3 (1 - 13)	3 (<1 - 29)	1 (<1 - 2)
Lead	µg/L	<1 (<1 - 2)	<1 (<1 - 20)	<1 (<1 - <1)
Zinc	µg/L	<10 (<10 - 24)	<10 (<10 - 80)	<10 (<10 - 12)
Flow	m ³ /s	0.020 (0.005 - 0.180)	0.035 (0.015 - 0.300)	0.030 (0.010 - 0.300)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Yuen Long Creek in 2018 (Part 1 of 2)

Parameter	Unit	<u>Yuen Long Creek</u>	
		YL1	YL2
Dissolved Oxygen	mg/L	4.3 (2.4 - 9.7)	3.5 (2.3 - 8.4)
pH		7.1 (6.7 - 7.6)	7.3 (7.0 - 7.4)
Suspended Solids	mg/L	15.5 (6.4 - 28.0)	7.3 (2.6 - 42.0)
5-Day Biochemical Oxygen Demand	mg/L	18.0 (3.2 - 37.0)	10.1 (4.3 - 27.0)
Chemical Oxygen Demand	mg/L	31 (7 - 75)	35 (24 - 92)
Oil & Grease	mg/L	0.8 (<0.5 - 2.8)	0.8 (<0.5 - 1.4)
<i>E. coli</i>	counts/ 100mL	120 000 (6 000 - 910 000)	87 000 (7 000 - 850 000)
Faecal Coliforms	counts/ 100mL	280 000 (22 000 - 4 000 000)	130 000 (10 000 - 1 100 000)
Ammonia-Nitrogen	mg/L	6.300 (0.810 - 38.000)	17.000 (7.200 - 25.000)
Nitrate-Nitrogen	mg/L	0.365 (<0.002 - 0.850)	0.245 (0.094 - 1.400)
Total Kjeldahl Nitrogen	mg/L	7.80 (3.60 - 43.00)	19.50 (8.40 - 27.00)
Orthophosphate Phosphorus	mg/L	0.765 (0.110 - 2.400)	2.150 (0.600 - 3.000)
Total Phosphorus	mg/L	0.97 (0.22 - 3.40)	2.65 (0.79 - 3.20)
Sulphide	mg/L	0.03 (<0.02 - 0.06)	0.05 (<0.02 - 0.13)
Aluminium	µg/L	54 (<50 - 146)	<50 (<50 - 168)
Cadmium	µg/L	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 1)	<1 (<1 - <1)
Copper	µg/L	2 (1 - 7)	3 (2 - 4)
Lead	µg/L	<1 (<1 - 1)	<1 (<1 - <1)
Zinc	µg/L	<10 (<10 - 37)	12 (<10 - 36)
Flow	m ³ /s	0.125 (0.097 - 0.525)	0.023 (0.010 - 0.075)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Yuen Long Creek in 2018 (Part 2 of 2)

Parameter	Unit	<u>Yuen Long Creek</u>	
		YL3	YL4
Dissolved Oxygen	mg/L	2.8 (1.6 - 5.9)	2.9 (1.7 - 4.9)
pH		7.3 (6.8 - 7.7)	6.9 (6.7 - 7.5)
Suspended Solids	mg/L	25.5 (21.0 - 64.0)	61.0 (18.0 - 130.0)
5-Day Biochemical Oxygen Demand	mg/L	68.0 (14.0 - 130.0)	140.0 (55.0 - 200.0)
Chemical Oxygen Demand	mg/L	92 (17 - 160)	165 (74 - 280)
Oil & Grease	mg/L	2.1 (<0.5 - 8.5)	5.6 (1.5 - 10.0)
<i>E. coli</i>	counts/ 100mL	1 000 000 (340 000 - 1 900 000)	1 700 000 (770 000 - 2 700 000)
Faecal Coliforms	counts/ 100mL	2 100 000 (720 000 - 7 800 000)	4 400 000 (1 900 000 - 11 000 000)
Ammonia-Nitrogen	mg/L	9.250 (2.000 - 17.000)	6.850 (3.400 - 9.700)
Nitrate-Nitrogen	mg/L	0.005 (<0.002 - 0.470)	<0.002 (<0.002 - 0.015)
Total Kjeldahl Nitrogen	mg/L	13.5 (4.50 - 22.00)	11.00 (8.10 - 15.00)
Orthophosphate Phosphorus	mg/L	0.855 (0.160 - 1.500)	0.285 (0.022 - 0.570)
Total Phosphorus	mg/L	1.45 (0.28 - 3.10)	1.10 (0.71 - 1.40)
Sulphide	mg/L	0.07 (0.03 - 0.18)	0.20 (0.07 - 0.33)
Aluminium	µg/L	<50 (<50 - 259)	<50 (<50 - 142)
Cadmium	µg/L	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 8)	<1 (<1 - 2)
Copper	µg/L	3 (1 - 12)	4 (2 - 16)
Lead	µg/L	<1 (<1 - 2)	<1 (<1 - 2)
Zinc	µg/L	<10 (<10 - 77)	11 (<10 - 79)
Flow	m ³ /s	0.400 (0.230 - 0.875)	0.135 (0.089 - 0.432)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Kam Tin River in 2018

Parameter	Unit	<u>Kam Tin River</u>	
		KT1	KT2
Dissolved Oxygen	mg/L	5.1 (2.6 - 7.3)	3.1 (1.5 - 7.3)
pH		7.1 (6.6 - 7.6)	7.3 (7.0 - 7.6)
Suspended Solids	mg/L	8.5 (4.6 - 31.0)	31.5 (7.8 - 67.0)
5-Day Biochemical Oxygen Demand	mg/L	8.5 (3.3 - 54.0)	26.5 (3.7 - 86.0)
Chemical Oxygen Demand	mg/L	20 (9 - 35)	50 (8 - 150)
Oil & Grease	mg/L	<0.5 (<0.5 - 0.6)	0.6 (<0.5 - 2.2)
<i>E. coli</i>	counts/ 100mL	71 000 (8 200 - 900 000)	82 000 (15 000 - 420 000)
Faecal Coliforms	counts/ 100mL	180 000 (37 000 - 1 300 000)	190 000 (26 000 - 1 100 000)
Ammonia-Nitrogen	mg/L	4.850 (1.100 - 14.000)	9.000 (1.000 - 15.000)
Nitrate-Nitrogen	mg/L	0.785 (<0.002 - 1.200)	0.275 (<0.002 - 0.660)
Total Kjeldahl Nitrogen	mg/L	6.60 (2.10 - 16.00)	12.00 (2.30 - 20.00)
Orthophosphate Phosphorus	mg/L	0.715 (0.210 - 2.200)	1.300 (0.200 - 3.100)
Total Phosphorus	mg/L	0.91 (0.35 - 2.90)	2.10 (0.35 - 3.80)
Sulphide	mg/L	0.03 (<0.02 - 0.06)	0.07 (<0.02 - 0.15)
Aluminium	µg/L	<50 (<50 - 81)	<50 (<50 - 127)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - 1)
Copper	µg/L	4 (2 - 94)	2 (<1 - 6)
Lead	µg/L	<1 (<1 - 4)	<1 (<1 - 1)
Zinc	µg/L	10 (<10 - 50)	<10 (<10 - 59)
Flow	m ³ /s	0.333 (0.198 - 0.588)	0.229 (0.078 - 0.525)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tin Shui Wai Nullah and Fairview Park Nullah in 2018

Parameter	Unit	<u>Tin Shui Wai Nullah</u>		<u>Fairview Park Nullah</u>
		TSR1	TSR2	FVR1
Dissolved Oxygen	mg/L	5.9 (2.4 - 8.9)	9.1 (8.0 - 12.4)	5.7 (4.1 - 8.8)
pH		7.3 (6.9 - 8.3)	7.6 (6.8 - 8.9)	7.5 (7.2 - 7.9)
Suspended Solids	mg/L	7.9 (1.7 - 32.0)	4.3 (2.2 - 9.5)	28.5 (8.2 - 58.0)
5-Day Biochemical Oxygen Demand	mg/L	9.7 (3.5 - 53.0)	1.9 (0.4 - 5.7)	6.5 (3.6 - 12.0)
Chemical Oxygen Demand	mg/L	22 (10 - 47)	9 (<2 - 15)	26 (12 - 48)
Oil & Grease	mg/L	<0.5 (<0.5 - 1.4)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	100 000 (10 000 - 520 000)	26 000 (6 000 - 130 000)	13 000 (1 700 - 82 000)
Faecal Coliforms	counts/ 100mL	390 000 (110 000 - 1 700 000)	50 000 (11 000 - 230 000)	32 000 (3 800 - 250 000)
Ammonia-Nitrogen	mg/L	1.500 (0.580 - 7.000)	0.315 (0.097 - 2.600)	2.400 (0.740 - 3.300)
Nitrate-Nitrogen	mg/L	0.760 (<0.002 - 1.300)	0.760 (0.480 - 1.200)	0.885 (0.360 - 1.900)
Total Kjeldahl Nitrogen	mg/L	3.00 (1.30 - 8.50)	0.98 (0.41 - 3.40)	3.60 (2.10 - 4.60)
Orthophosphate Phosphorus	mg/L	0.165 (0.007 - 0.540)	0.052 (0.004 - 0.130)	0.310 (0.012 - 0.520)
Total Phosphorus	mg/L	0.27 (0.14 - 0.93)	0.07 (0.03 - 0.17)	0.57 (0.33 - 0.85)
Sulphide	mg/L	<0.02 (<0.02 - 0.06)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.04)
Aluminium	µg/L	82 (<50 - 159)	133 (<50 - 434)	<50 (<50 - 296)
Cadmium	µg/L	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - <1)	1 (<1 - 3)
Copper	µg/L	3 (<1 - 11)	1 (<1 - 4)	2 (<1 - 5)
Lead	µg/L	<1 (<1 - 2)	<1 (<1 - 1)	<1 (<1 - 3)
Zinc	µg/L	19 (<10 - 68)	<10 (<10 - 29)	13 (<10 - 48)
Flow	m ³ /s	NM	0.113 (0.025 - 0.540)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Ha Pak Nai Stream, Pak Nai Stream and Sheung Pak Nai Stream in 2018

Parameter	Unit	Ha Pak Nai Stream	Pak Nai Stream	Sheung Pak Nai Stream
		DB1	DB3	DB5
Dissolved Oxygen	mg/L	8.4 (7.6 - 10.1)	8.1 (7.4 - 9.8)	8.0 (7.1 - 9.7)
pH		6.7 (5.8 - 7.1)	6.4 (5.5 - 6.8)	6.7 (6.0 - 7.1)
Suspended Solids	mg/L	1.8 (1.2 - 1 900.0)	2.5 (1.1 - 1 200.0)	5.1 (1.2 - 780.0)
5-Day Biochemical Oxygen Demand	mg/L	0.2 (<0.1 - 3.6)	0.2 (<0.1 - 2.8)	1.7 (0.5 - 7.2)
Chemical Oxygen Demand	mg/L	3 (<2 - 33)	5 (<2 - 39)	6 (4 - 15)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.6)
<i>E. coli</i>	counts/ 100mL	160 (12 - 6 300)	440 (52 - 6 800)	19 000 (1 200 - 420 000)
Faecal Coliforms	counts/ 100mL	1 500 (330 - 27 000)	2 500 (260 - 61 000)	27 000 (1 600 - 440 000)
Ammonia-Nitrogen	mg/L	0.017 (<0.005 - 0.240)	0.026 (<0.005 - 0.140)	0.335 (0.032 - 2.500)
Nitrate-Nitrogen	mg/L	0.340 (0.170 - 1.300)	0.300 (0.170 - 0.590)	0.270 (0.210 - 0.320)
Total Kjeldahl Nitrogen	mg/L	0.17 (0.06 - 3.80)	0.21 (<0.05 - 3.90)	1.00 (0.26 - 4.80)
Orthophosphate Phosphorus	mg/L	0.003 (<0.002 - 0.006)	0.004 (<0.002 - 0.008)	0.104 (0.003 - 0.390)
Total Phosphorus	mg/L	<0.02 (<0.02 - 0.19)	<0.02 (<0.02 - 0.24)	0.14 (<0.02 - 0.57)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	315 (62 - 14 140)	295 (<50 - 19 524)	229 (<50 - 9 382)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	<1 (<1 - 2)	<1 (<1 - 2)	1 (<1 - 2)
Lead	µg/L	<1 (<1 - 5)	<1 (<1 - 5)	<1 (<1 - 6)
Zinc	µg/L	<10 (<10 - 11)	<10 (<10 - 19)	<10 (<10 - 41)
Flow	m ³ /s	0.028 (0.005 - 0.300)	0.050 (0.010 - 0.140)	0.060 (0.015 - 0.480)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Ngau Hom Sha Stream, Tai Shui Hang Stream and Tsang Kok Stream in 2018

Parameter	Unit	<u>Ngau Hom Sha Stream</u>	<u>Tai Shui Hang Stream</u>	<u>Tsang Kok Stream</u>
		DB6	DB2	DB8
Dissolved Oxygen	mg/L	8.3 (6.2 - 9.8)	8.3 (7.1 - 10.2)	8.0 (5.5 - 10.8)
pH		6.7 (6.2 - 7.1)	7.3 (6.2 - 7.5)	7.4 (6.2 - 8.3)
Suspended Solids	mg/L	4.4 (1.4 - 22.0)	2.4 (1.0 - 440.0)	9.2 (2.2 - 850.0)
5-Day Biochemical Oxygen Demand	mg/L	0.5 (<0.1 - 1.2)	0.5 (<0.1 - 8.0)	1.2 (0.1 - 25.0)
Chemical Oxygen Demand	mg/L	4 (<2 - 8)	4 (<2 - 19)	8 (3 - 180)
Oil & Grease	mg/L	<0.5 (<0.5 - 0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.6)
<i>E. coli</i>	counts/ 100mL	190 (34 - 1 500)	300 (26 - 32 000)	830 (210 - 9 600)
Faecal Coliforms	counts/ 100mL	2 000 (330 - 12 000)	2 200 (260 - 120 000)	7 000 (2 200 - 39 000)
Ammonia-Nitrogen	mg/L	0.078 (0.025 - 0.420)	0.026 (<0.005 - 4.400)	0.071 (0.006 - 23.000)
Nitrate-Nitrogen	mg/L	0.230 (0.023 - 0.370)	0.205 (0.120 - 0.420)	1.500 (<0.002 - 2.200)
Total Kjeldahl Nitrogen	mg/L	0.27 (0.08 - 0.61)	0.16 (0.08 - 4.80)	0.48 (0.12 - 27.00)
Orthophosphate Phosphorus	mg/L	0.056 (0.004 - 0.260)	0.004 (<0.002 - 0.029)	0.008 (0.005 - 0.170)
Total Phosphorus	mg/L	0.09 (0.04 - 0.35)	<0.02 (<0.02 - 0.04)	0.02 (<0.02 - 0.64)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	88 (<50 - 700)	164 (<50 - 6 949)	146 (<50 - 5 927)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - 1)
Copper	µg/L	<1 (<1 - 1)	1 (<1 - 2)	1 (<1 - 8)
Lead	µg/L	<1 (<1 - 1)	<1 (<1 - 3)	<1 (<1 - 6)
Zinc	µg/L	<10 (<10 - 18)	<10 (<10 - 12)	<10 (<10 - 24)
Flow	m ³ /s	0.030 (0.008 - 0.125)	0.200 (0.010 - 3.200)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Mui Wo River in 2018 (Part 1 of 2)

Parameter	Unit	Mui Wo River		
		MW1	MW2	MW3
Dissolved Oxygen	mg/L	7.9 (6.5 - 10.1)	7.6 (5.6 - 10.8)	8.2 (6.8 - 10.3)
pH		7.4 (6.2 - 8.0)	7.3 (6.5 - 8.0)	6.3 (5.8 - 7.3)
Suspended Solids	mg/L	2.1 (0.5 - 7.1)	3.4 (2.1 - 8.0)	1.1 (0.6 - 3.7)
5-Day Biochemical Oxygen Demand	mg/L	0.5 (<0.1 - 1.1)	1.1 (0.3 - 2.9)	0.3 (<0.1 - 0.8)
Chemical Oxygen Demand	mg/L	6 (<2 - 9)	9 (4 - 27)	3 (<2 - 8)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	570 (140 - 5 700)	1 900 (45 - 22 000)	120 (17 - 1 200)
Faecal Coliforms	counts/ 100mL	3 900 (500 - 47 000)	7 100 (160 - 73 000)	2 000 (400 - 30 000)
Ammonia-Nitrogen	mg/L	0.061 (0.011 - 0.450)	0.330 (0.026 - 1.100)	0.016 (0.007 - 0.026)
Nitrate-Nitrogen	mg/L	0.380 (0.190 - 0.990)	0.275 (0.130 - 0.580)	0.440 (0.190 - 0.650)
Total Kjeldahl Nitrogen	mg/L	0.22 (0.11 - 0.54)	0.49 (0.19 - 1.60)	0.15 (0.07 - 0.32)
Orthophosphate Phosphorus	mg/L	0.067 (0.011 - 0.160)	0.047 (0.021 - 0.230)	0.064 (0.013 - 0.095)
Total Phosphorus	mg/L	0.10 (0.04 - 0.20)	0.09 (0.04 - 0.35)	0.07 (<0.02 - 0.12)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	53 (<50 - 124)	<50 (<50 - 270)	77 (<50 - 185)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 4)	<1 (<1 - 5)	<1 (<1 - 1)
Copper	µg/L	1 (<1 - 5)	1 (<1 - 8)	<1 (<1 - 2)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	<10 (<10 - 19)	<10 (<10 - 19)	<10 (<10 - 18)
Flow	m ³ /s	0.060 (0.040 - 0.300)	NM	0.015 (0.005 - 0.600)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Mui Wo River in 2018 (Part 2 of 2)

Parameter	Unit	<u>Mui Wo River</u>	
		MW4	MW5
Dissolved Oxygen	mg/L	7.0 (5.2 - 9.2)	7.6 (5.6 - 9.5)
pH		7.2 (6.7 - 8.0)	7.2 (6.7 - 8.0)
Suspended Solids	mg/L	7.6 (2.7 - 24.0)	5.4 (1.1 - 37.0)
5-Day Biochemical Oxygen Demand	mg/L	0.8 (0.3 - 2.7)	1.7 (0.3 - 3.0)
Chemical Oxygen Demand	mg/L	10 (5 - 26)	12 (2 - 25)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	900 (39 - 6 700)	4 000 (1 000 - 17 000)
Faecal Coliforms	counts/ 100mL	2 700 (480 - 8 300)	22 000 (3 300 - 100 000)
Ammonia-Nitrogen	mg/L	0.265 (0.083 - 1.400)	0.765 (0.110 - 1.900)
Nitrate-Nitrogen	mg/L	0.245 (0.180 - 0.570)	0.240 (0.140 - 0.370)
Total Kjeldahl Nitrogen	mg/L	0.48 (0.23 - 1.90)	1.05 (0.23 - 2.40)
Orthophosphate Phosphorus	mg/L	0.044 (0.016 - 0.067)	0.068 (0.027 - 0.190)
Total Phosphorus	mg/L	0.09 (0.05 - 0.26)	0.16 (0.04 - 0.43)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 263)	54 (<50 - 236)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	2 (<1 - 5)	<1 (<1 - 2)
Copper	µg/L	2 (<1 - 8)	1 (<1 - 3)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	<10 (<10 - 23)	<10 (<10 - 35)
Flow	m ³ /s	0.200 (0.050 - 0.600)	0.140 (0.016 - 0.240)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tung Chung River in 2018

Parameter	Unit	Tung Chung River		
		TC1	TC2	TC3
Dissolved Oxygen	mg/L	7.9 (5.5 - 9.4)	8.7 (7.6 - 10.8)	8.3 (6.6 - 9.5)
pH		6.6 (6.1 - 7.0)	7.9 (6.8 - 10.0)	7.3 (6.3 - 8.1)
Suspended Solids	mg/L	1.0 (<0.5 - 7.9)	3.3 (1.3 - 8.2)	2.5 (1.0 - 7.7)
5-Day Biochemical Oxygen Demand	mg/L	0.3 (<0.1 - 0.9)	1.0 (<0.1 - 2.7)	5.4 (0.5 - 13.0)
Chemical Oxygen Demand	mg/L	3 (<2 - 8)	4 (<2 - 13)	8 (2 - 16)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.8)
<i>E. coli</i>	counts/ 100mL	59 (19 - 170)	180 (8 - 820)	16 000 (2 200 - 55 000)
Faecal Coliforms	counts/ 100mL	930 (200 - 6 200)	2 000 (390 - 8 200)	52 000 (11 000 - 210 000)
Ammonia-Nitrogen	mg/L	0.012 (<0.005 - 0.052)	0.028 (0.006 - 0.062)	1.300 (0.017 - 2.300)
Nitrate-Nitrogen	mg/L	0.057 (<0.002 - 0.210)	0.059 (<0.002 - 0.230)	0.096 (0.061 - 0.160)
Total Kjeldahl Nitrogen	mg/L	0.11 (<0.05 - 0.19)	0.19 (0.07 - 0.36)	1.75 (0.24 - 2.90)
Orthophosphate Phosphorus	mg/L	0.003 (<0.002 - 0.029)	0.007 (0.003 - 0.015)	0.064 (0.007 - 0.200)
Total Phosphorus	mg/L	<0.02 (<0.02 - 0.04)	<0.02 (<0.02 - 0.04)	0.13 (<0.02 - 0.29)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 88)	<50 (<50 - 66)	<50 (<50 - 53)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - 1)	<1 (<1 - <1)
Copper	µg/L	<1 (<1 - 2)	<1 (<1 - 1)	1 (<1 - 2)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	<10 (<10 - 10)	<10 (<10 - 15)	<10 (<10 - 22)
Flow	m ³ /s	0.045 (0.010 - 0.300)	0.090 (0.030 - 0.480)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tuen Mun River in 2018 (Part 1 of 2)

Parameter	Unit	Tuen Mun River		
		TN1	TN2	TN3
Dissolved Oxygen	mg/L	4.8 (2.9 - 6.1)	7.7 (5.6 - 9.0)	4.9 (3.1 - 7.1)
pH		7.4 (7.1 - 7.9)	7.2 (6.6 - 7.9)	7.5 (7.2 - 7.9)
Suspended Solids	mg/L	10.5 (3.0 - 22.0)	8.4 (2.5 - 26.0)	6.5 (3.2 - 23.0)
5-Day Biochemical Oxygen Demand	mg/L	22.0 (10.0 - 45.0)	3.5 (0.7 - 28.0)	2.6 (1.0 - 13.0)
Chemical Oxygen Demand	mg/L	26 (15 - 54)	10 (3 - 33)	15 (7 - 23)
Oil & Grease	mg/L	0.8 (<0.5 - 2.1)	<0.5 (<0.5 - 0.8)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	130 000 (34 000 - 260 000)	53 000 (12 000 - 200 000)	11 000 (610 - 100 000)
Faecal Coliforms	counts/ 100mL	460 000 (120 000 - 1 000 000)	89 000 (26 000 - 380 000)	86 000 (3 800 - 720 000)
Ammonia-Nitrogen	mg/L	6.650 (3.800 - 11.000)	1.700 (0.510 - 5.700)	0.420 (0.270 - 0.820)
Nitrate-Nitrogen	mg/L	0.350 (<0.002 - 2.200)	2.000 (0.700 - 3.500)	0.520 (0.220 - 0.860)
Total Kjeldahl Nitrogen	mg/L	9.35 (5.50 - 13.00)	2.20 (0.78 - 8.00)	0.80 (0.46 - 1.50)
Orthophosphate Phosphorus	mg/L	0.525 (0.310 - 0.940)	0.180 (0.007 - 0.600)	0.037 (0.002 - 0.084)
Total Phosphorus	mg/L	0.87 (0.54 - 1.20)	0.28 (0.06 - 0.87)	0.07 (0.04 - 0.16)
Sulphide	mg/L	0.04 (<0.02 - 0.08)	<0.02 (<0.02 - 0.05)	<0.02 (<0.02 - 0.03)
Aluminium	µg/L	<50 (<50 - 113)	122 (<50 - 614)	54 (<50 - 128)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - 2)	2 (1 - 5)
Copper	µg/L	2 (1 - 5)	1 (<1 - 4)	4 (1 - 7)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - 3)	<1 (<1 - <1)
Zinc	µg/L	13 (<10 - 38)	<10 (<10 - 36)	10 (<10 - 28)
Flow	m ³ /s	0.154 (0.075 - 0.240)	0.030 (0.010 - 0.216)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tuen Mun River in 2018 (Part 2 of 2)

Parameter	Unit	Tuen Mun River		
		TN4	TN5	TN6
Dissolved Oxygen	mg/L	5.4 (3.1 - 10.3)	5.4 (3.1 - 7.1)	5.4 (3.3 - 6.8)
pH		7.6 (7.2 - 8.1)	7.5 (7.2 - 7.9)	7.4 (7.2 - 7.7)
Suspended Solids	mg/L	6.6 (3.2 - 150.0)	6.4 (2.6 - 230.0)	5.6 (2.4 - 14.0)
5-Day Biochemical Oxygen Demand	mg/L	1.7 (0.9 - 6.6)	2.0 (1.2 - 28.0)	2.5 (1.2 - 4.4)
Chemical Oxygen Demand	mg/L	12 (6 - 24)	14 (7 - 46)	12 (6 - 21)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	6 700 (600 - 71 000)	8 100 (500 - 86 000)	11 000 (1 400 - 160 000)
Faecal Coliforms	counts/ 100mL	46 000 (6 100 - 340 000)	47 000 (4 400 - 770 000)	62 000 (6 900 - 950 000)
Ammonia-Nitrogen	mg/L	0.580 (0.190 - 0.990)	0.460 (0.230 - 0.800)	0.480 (0.200 - 1.100)
Nitrate-Nitrogen	mg/L	0.490 (0.240 - 1.000)	0.540 (<0.002 - 0.920)	0.525 (0.150 - 0.870)
Total Kjeldahl Nitrogen	mg/L	0.73 (0.44 - 1.80)	0.92 (0.50 - 2.00)	0.79 (0.50 - 1.80)
Orthophosphate Phosphorus	mg/L	0.042 (0.002 - 0.098)	0.039 (0.003 - 0.086)	0.048 (0.012 - 0.090)
Total Phosphorus	mg/L	0.08 (0.04 - 0.18)	0.09 (0.04 - 0.34)	0.08 (0.04 - 0.16)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.39)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	74 (<50 - 193)	66 (<50 - 152)	53 (<50 - 149)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.1)
Chromium	µg/L	2 (1 - 4)	3 (1 - 4)	2 (1 - 4)
Copper	µg/L	4 (1 - 13)	4 (1 - 8)	4 (3 - 13)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	11 (<10 - 21)	11 (<10 - 27)	12 (<10 - 33)
Flow	m ³ /s	NM	NM	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Pai Min Kok Stream and Kau Wa Keng Stream in 2018

Parameter	Unit	Pai Min Kok Stream		Kau Wa Keng Stream
		AN1	AN2	KW3
Dissolved Oxygen	mg/L	8.3 (7.8 - 10.2)	9.5 (7.9 - 11.6)	8.5 (7.7 - 10.1)
pH		8.0 (7.6 - 8.3)	8.1 (7.3 - 9.2)	7.4 (7.2 - 8.4)
Suspended Solids	mg/L	5.2 (1.2 - 12.0)	1.8 (1.0 - 3.4)	4.1 (1.3 - 180.0)
5-Day Biochemical Oxygen Demand	mg/L	0.9 (0.3 - 9.0)	0.5 (<0.1 - 4.2)	2.7 (0.8 - 7.7)
Chemical Oxygen Demand	mg/L	11 (5 - 20)	5 (3 - 12)	12 (4 - 30)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 1.0)
<i>E. coli</i>	counts/ 100mL	8 600 (400 - 83 000)	6 700 (300 - 70 000)	37 000 (36 - 250 000)
Faecal Coliforms	counts/ 100mL	25 000 (3 200 - 240 000)	18 000 (4 800 - 140 000)	68 000 (47 - 310 000)
Ammonia-Nitrogen	mg/L	0.062 (0.018 - 0.940)	0.023 (0.010 - 2.000)	0.460 (0.086 - 1.300)
Nitrate-Nitrogen	mg/L	0.625 (0.400 - 1.300)	0.385 (0.190 - 0.620)	1.950 (1.400 - 2.500)
Total Kjeldahl Nitrogen	mg/L	0.39 (0.20 - 1.50)	0.29 (0.11 - 2.70)	0.97 (0.56 - 2.10)
Orthophosphate Phosphorus	mg/L	0.049 (0.007 - 0.250)	0.044 (0.026 - 0.150)	0.060 (0.005 - 0.210)
Total Phosphorus	mg/L	0.09 (0.04 - 0.31)	0.07 (0.03 - 0.16)	0.12 (0.05 - 0.40)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.04)
Aluminium	µg/L	107 (<50 - 470)	317 (<50 - 685)	<50 (<50 - 2 821)
Cadmium	µg/L	<0.1 (<0.1 - 0.3)	<0.1 (<0.1 - <0.1)	1.1 (<0.1 - 5.1)
Chromium	µg/L	1 (<1 - 3)	<1 (<1 - <1)	<1 (<1 - 2)
Copper	µg/L	4 (2 - 12)	3 (2 - 4)	2 (1 - 7)
Lead	µg/L	<1 (<1 - 3)	<1 (<1 - <1)	<1 (<1 - 3)
Zinc	µg/L	15 (<10 - 59)	10 (<10 - 15)	103 (<10 - 266)
Flow	m ³ /s	NM	0.009 (0.002 - 0.022)	0.035 (0.010 - 0.120)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
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Summary of water quality monitoring data for Sam Dip Tam Stream in 2018

Parameter	Unit	Sam Dip Tam Stream		
		TW1	TW2	TW3
Dissolved Oxygen	mg/L	8.3 (7.8 - 10.4)	8.8 (7.9 - 10.5)	8.6 (7.9 - 10.6)
pH		7.5 (7.1 - 8.0)	7.9 (7.5 - 8.2)	7.9 (7.5 - 8.4)
Suspended Solids	mg/L	1.4 (0.6 - 3.5)	1.5 (0.7 - 18.0)	1.4 (0.7 - 2.5)
5-Day Biochemical Oxygen Demand	mg/L	0.6 (0.3 - 1.4)	0.8 (0.5 - 9.4)	0.9 (0.3 - 2.7)
Chemical Oxygen Demand	mg/L	5 (<2 - 9)	5 (3 - 10)	4 (<3 - 15)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	3 400 (230 - 59 000)	5 200 (100 - 33 000)	8 000 (350 - 2 800 000)
Faecal Coliforms	counts/ 100mL	15 000 (2 600 - 78 000)	20 000 (500 - 84 000)	31 000 (4 300 - 2 800 000)
Ammonia-Nitrogen	mg/L	0.023 (0.009 - 0.053)	0.130 (0.041 - 0.420)	0.075 (0.049 - 0.790)
Nitrate-Nitrogen	mg/L	0.820 (0.650 - 1.100)	1.150 (1.000 - 1.400)	1.450 (1.200 - 1.600)
Total Kjeldahl Nitrogen	mg/L	0.29 (0.17 - 0.43)	0.40 (0.19 - 0.96)	0.40 (0.29 - 1.20)
Orthophosphate Phosphorus	mg/L	0.042 (0.009 - 0.061)	0.098 (0.020 - 0.140)	0.100 (0.050 - 0.240)
Total Phosphorus	mg/L	0.06 (0.04 - 0.08)	0.12 (0.06 - 0.20)	0.12 (0.08 - 0.29)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - <50)	<50 (<50 - 68)	<50 (<50 - 108)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	<1 (<1 - 2)	2 (<1 - 2)	2 (<1 - 3)
Lead	µg/L	<1 (<1 - 1)	<1 (<1 - 1)	<1 (<1 - 2)
Zinc	µg/L	<10 (<10 - 18)	<10 (<10 - 36)	<10 (<10 - 29)
Flow	m ³ /s	NM	0.083 (0.038 - 0.260)	NM

- Notes:
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Summary of water quality monitoring data for Kai Tak River in 2018 (Part 1 of 2)

Parameter	Unit	Kai Tak River		
		KN1	KN2	KN3
Dissolved Oxygen	mg/L	5.1 (3.7 - 7.3)	6.2 (5.2 - 7.0)	6.7 (6.4 - 7.3)
pH		7.2 (7.0 - 7.4)	7.2 (7.1 - 7.3)	7.3 (7.1 - 7.4)
Suspended Solids	mg/L	2.5 (0.7 - 6.7)	4.4 (2.9 - 13.0)	6.6 (3.4 - 30.0)
5-Day Biochemical Oxygen Demand	mg/L	3.0 (1.2 - 4.5)	3.7 (1.6 - 5.0)	4.3 (1.4 - 14.0)
Chemical Oxygen Demand	mg/L	21 (11 - 40)	31 (12 - 46)	29 (16 - 39)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	22 000 (1 000 - 170 000)	29 000 (3 000 - 220 000)	26 000 (2 900 - 230 000)
Faecal Coliforms	counts/ 100mL	73 000 (2 400 - 660 000)	65 000 (5 900 - 340 000)	56 000 (5 100 - 500 000)
Ammonia-Nitrogen	mg/L	1.450 (0.620 - 4.700)	0.990 (0.490 - 3.900)	0.790 (0.340 - 3.900)
Nitrate-Nitrogen	mg/L	3.450 (1.600 - 4.900)	3.800 (2.800 - 6.600)	4.300 (2.500 - 7.800)
Total Kjeldahl Nitrogen	mg/L	2.40 (1.00 - 6.00)	2.00 (1.40 - 5.30)	1.70 (1.50 - 5.30)
Orthophosphate Phosphorus	mg/L	1.150 (0.300 - 1.700)	1.000 (0.620 - 2.000)	1.100 (0.480 - 1.900)
Total Phosphorus	mg/L	1.40 (0.35 - 1.70)	1.30 (0.75 - 2.00)	1.20 (0.59 - 2.00)
Sulphide	mg/L	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - 0.04)
Aluminium	µg/L	<50 (<50 - 71)	<50 (<50 - <50)	<50 (<50 - <50)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.2)	<0.1 (<0.1 - 0.3)
Chromium	µg/L	2 (<1 - 3)	2 (<1 - 3)	2 (<1 - 2)
Copper	µg/L	3 (2 - 5)	3 (2 - 5)	3 (2 - 4)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - 1)
Zinc	µg/L	15 (12 - 29)	15 (11 - 24)	16 (14 - 40)
Flow	m ³ /s	NM	NM	NM

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Summary of water quality monitoring data for Kai Tak River in 2018 (Part 2 of 2)

Parameter	Unit	Kai Tak River		
		KN4	KN5	KN7
Dissolved Oxygen	mg/L	6.8 (6.5 - 7.0)	7.2 (6.8 - 8.5)	7.4 (7.1 - 8.6)
pH		7.2 (6.9 - 7.4)	7.1 (6.9 - 7.2)	6.9 (6.9 - 7.2)
Suspended Solids	mg/L	7.3 (3.2 - 100.0)	7.3 (4.3 - 13.0)	5.9 (1.9 - 12.0)
5-Day Biochemical Oxygen Demand	mg/L	5.3 (2.2 - 51.0)	4.7 (3.0 - 11.0)	4.0 (1.7 - 11.0)
Chemical Oxygen Demand	mg/L	32 (12 - 49)	33 (13 - 56)	31 (12 - 40)
Oil & Grease	mg/L	<0.5 (<0.5 - 1.2)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	24 000 (2 300 - 270 000)	22 000 (1 800 - 210 000)	2 000 (740 - 5 000)
Faecal Coliforms	counts/ 100mL	52 000 (3 700 - 1 200 000)	49 000 (4 400 - 350 000)	4 700 (1 200 - 10 000)
Ammonia-Nitrogen	mg/L	0.725 (0.044 - 3.200)	0.490 (0.240 - 2.000)	0.380 (0.210 - 1.800)
Nitrate-Nitrogen	mg/L	4.700 (0.350 - 5.700)	5.000 (3.000 - 6.800)	5.250 (3.200 - 6.700)
Total Kjeldahl Nitrogen	mg/L	1.95 (1.40 - 4.80)	1.75 (1.20 - 3.90)	1.70 (1.20 - 3.60)
Orthophosphate Phosphorus	mg/L	1.150 (0.250 - 1.800)	1.150 (0.650 - 1.800)	1.200 (0.650 - 1.800)
Total Phosphorus	mg/L	1.35 (0.66 - 2.00)	1.45 (0.81 - 1.90)	1.40 (0.81 - 2.00)
Sulphide	mg/L	0.03 (<0.02 - 0.12)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - <50)	<50 (<50 - 114)	<50 (<50 - <50)
Cadmium	µg/L	<0.1 (<0.1 - 0.3)	<0.1 (<0.1 - 0.4)	<0.1 (<0.1 - 0.4)
Chromium	µg/L	2 (<1 - 3)	2 (<1 - 2)	2 (<1 - 3)
Copper	µg/L	3 (2 - 5)	4 (2 - 5)	3 (2 - 6)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	17 (13 - 36)	18 (13 - 28)	20 (12 - 40)
Flow	m ³ /s	NM	11.300 (6.700 - 17.150)	1.283 (0.150 - 8.250)

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Compliance with the river Water Quality Objectives in 2018

Watercourse	pH	5-Day Biochemical Oxygen Demand	Chemical Oxygen Demand	Dissolved Oxygen	Suspended Solids*	Overall Compliance
Eastern New Territories						
Shing Mun River	87%	91%	93%	100%	100%	94%
Lam Tsuen River	97%	93%	93%	100%	100%	96%
Tai Po River	92%	100%	100%	100%	100%	98%
Tai Po Kau Stream	100%	100%	100%	100%	100%	100%
Shan Liu Stream	100%	100%	100%	100%	100%	100%
Tung Tze Stream	100%	100%	92%	100%	100%	98%
Ho Chung River	100%	100%	100%	100%	100%	100%
Sha Kok Mei Stream	100%	92%	100%	100%	100%	98%
Tai Chung Hau Stream	100%	96%	100%	100%	100%	99%
Tseng Lan Shue Stream	100%	58%	92%	97%	100%	89%
Northwestern New Territories						
River Indus	86%	56%	61%	97%	67%	73%
River Beas	94%	31%	61%	100%	100%	77%
River Ganges	97%	56%	72%	94%	100%	84%
Yuen Long Creek	100%	0%	6%	35%	50%	38%
Kam Tin River	100%	0%	21%	63%	50%	47%
Tin Shui Wai Nullah	100%	58%	88%	92%	100%	88%
Fairview Park Nullah	100%	33%	67%	100%	0%	60%
Ha Pak Nai Stream	92%	100%	92%	100%	100%	97%
Tai Shui Hang Stream	100%	92%	100%	100%	100%	98%
Pak Nai Stream	83%	100%	92%	100%	100%	95%
Sheung Pak Nai Stream	100%	83%	100%	100%	100%	97%
Ngau Hom Sha Stream	100%	100%	100%	100%	100%	100%
Tsang Kok Stream	100%	83%	83%	100%	100%	93%
Lantau Island						
Mui Wo River	80%	100%	100%	100%	100%	96%
Tung Chung River	97%	81%	100%	100%	100%	96%
Southwestern New Territories & Kowloon						
Tuen Mun River	100%	65%	86%	86%	100%	88%
Pai Min Kok Stream	92%	96%	100%	100%	100%	98%
Kau Wa Keng Stream	100%	83%	100%	100%	100%	97%
Sam Dip Tam Stream	100%	97%	100%	100%	100%	99%
Kai Tak River	100%	70%	53%	99%	100%	84%
Average Compliance (All monitoring stations)	95%	75%	82%	94%	94%	88%

* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Water Quality Index (WQI) for river waters of Hong Kong

The Water Quality Index (WQI) is a numerical value used to denote, based on monitoring data, the general ecological health of a river. It is relevant to conserving the primary beneficial use of rivers for maintenance of aquatic life. The WQI is calculated based on three key parameters: dissolved oxygen, 5-day biochemical oxygen demand (BOD₅) and ammonia-nitrogen content. The measurements of these parameters are evaluated using the following table:

Calculation of Water Quality Index

No. of points awarded	Dissolved Oxygen (% saturation)	BOD ₅ (mg/L)	Ammonia-Nitrogen (mg/L)
1	91 – 110	< 3	< 0.5
2	71 – 90 111 – 120	3.1 – 6.0	0.5 – 1.0
3	51 – 70 121 – 130	6.1 – 9.0	1.1 – 2.0
4	31 – 50	9.1 – 15.0	2.1 – 5.0
5	< 30 or > 130	> 15.0	> 5.0

All parameters carry equal weighting and the monthly index is the sum of points of the three parameters. The annual WQI of a river station is the average of all monthly WQI in the year. The WQI values range from 3 to 15, corresponding to the following water quality conditions:

Water Quality Index gradings

WQI	Water quality condition
3.0 – 4.5	Excellent
4.6 – 7.5	Good
7.6 – 10.5	Fair
10.6 – 13.5	Bad
13.6 – 15.0	Very Bad

Summary of river water quality improvements in 18 districts in 2018

District	Improvement	Remaining Problem
New Territories		
Kwai Tsing	All factory discharges are now controlled under the Water Pollution Control Ordinance. Domestic and commercial premises are connected to government sewers. Water quality in Kau Wa Keng Stream improved from “Bad” in 1991 to “Excellent” in 2018. Its <i>E. coli</i> level has been reduced by over 60% compared with 1998. Sewerage works for villages are being planned.	Still affected by expedient connections and a small number of unsewered villages
Tsuen Wan	All factory discharges are now controlled under the Water Pollution Control Ordinance. Domestic and commercial premises are connected to government sewers. Livestock farm discharges in the area have been eliminated under the Livestock Waste Control Scheme. Sam Dip Tam Stream improved from “Fair” to “Good” in 1988 to “Excellent” in 2018. Its <i>E. coli</i> level has been reduced by 80% compared with 1999. Pai Min Kok Stream improved from “Bad” in 1988 to “Excellent” in 2018. Its <i>E. coli</i> level has been reduced by over 90% compared with 1989. Sewerage works at hinterland of affected beaches have been completed and progressive connection of unsewered properties to the new public sewers is being made. Sewer provisions for other villages in the area are being planned.	Still affected by expedient connections and a small number of unsewered villages.
Tuen Mun	All factory discharges are now controlled under the Water Pollution Control Ordinance. Domestic and commercial premises are connected to government sewers. Livestock farm discharges are now controlled under the Livestock Waste Control Scheme. Tuen Mun River upstream improved from “Very Bad” in 1988 to “Good” to “Bad” in 2018. Its <i>E. coli</i> level has been reduced by over 90% compared with 1988. Tuen Mun River mid-stream and downstream improved from “Bad” in 1988 to “Good” in 2018. The <i>E. coli</i> level has been reduced by over 90% compared with 1988. Sewerage works for villages are progressively being planned and implemented. The Pillar Point Sewage Treatment Works have been upgraded to chemically enhanced primary treatment level with UV disinfection since May 2014. A new trunk sewer to the west of Tuen Mun River was completed in 2015.	Still affected by expedient connections and unsewered villages especially in the upstream region.
Yuen Long	Factory discharges are controlled under the Water Pollution Control Ordinance. Most of the domestic and commercial premises in the urbanised town center are connected to government sewers. Under the Livestock Waste Control Scheme, most of the farms had ceased operation. From “Very Bad” in 1986, Yuen Long Creek upstream and Kam Tin River both changed to “Fair” to “Bad” in 2018 while Yuen Long Creek downstream was “Very Bad” in 2018 due to sewer incidents. The <i>E. coli</i> level of these rivers has been reduced by around 70% compared with 1998. Tin Shui Wai Nullah improved from “Fair” to “Bad” in 1993 to “Fair” to “Good” in 2018. The <i>E. coli</i> level of this nullah has been reduced by over 90% compared with 1992. Trunk sewers, pumping stations, and village sewerage are being planned and constructed progressively. The sewage treatment works at San Wai and Yuen Long will be expanded in anticipation of projected population growth, and upgraded to produce higher quality effluent.	Still affected by polluted discharges from unsewered villages and expedient discharges from industrial enterprises and livestock farms.

District	Improvement	Remaining Problem
New Territories		
North	Factory discharges are controlled under the Water Pollution Control Ordinance. Most of the domestic and commercial premises in the urbanised town centers are connected to government sewers. Under the Livestock Waste Control Scheme, most farms have ceased operation. The downstream stations of River Indus, River Beas and River Ganges improved from “Very Bad” in 1987 to “Fair” to “Good” in 2018. The <i>E. coli</i> level of these rivers has been reduced by over 90% compared with 1990. Sewerage works for villages are progressively being planned and implemented. The sewage treatment works at Sha Tau Kok and Shek Wu Hui Sewage Treatment Works will be expanded in anticipation of projected population growth and establishment of the New Development Areas and upgraded to produce higher quality effluent.	Still affected by polluted discharges from unsewered villages and expedient discharges from industrial enterprises and livestock farms.
Tai Po	Factory discharges are controlled under the Water Pollution Control Ordinance. Most of the domestic and commercial premises in the urbanised town centers are connected to government sewers. Under the Livestock Waste Control Scheme, livestock farms have ceased operation. The water quality of downstream stations of rivers in the area improved from involving “Very Bad” in 1986 to “Good” to “Excellent” in 2018. The <i>E. coli</i> level of the downstream stations of both Lam Tsuen River and Tai Po River has been reduced by over 85% compared with 1999. Sewerage works for villages are progressively being planned and implemented. The Tai Po Sewage Treatment Works acquired UV disinfection facilities in 2010 and its treatment capacity has been increased to 120 000 m ³ /d since 2015.	Still affected by polluted discharges from a small number of unsewered villages and expedient connections.
Sha Tin	Factory discharges are controlled under the Water Pollution Control Ordinance. Most of the domestic and commercial premises in the urbanised town center are connected to government sewers. Under the Livestock Waste Control Scheme, livestock farms have ceased operation. The water quality of the Shing Mun River main channel improved from “Fair” in 1986 to “Excellent” in 2018. The <i>E. coli</i> level of the main channel has been reduced by over 90% compared with 1988. Village sewerage is being extended to most of the remaining villages in the area. The Sha Tin Sewage Treatment Works acquired UV disinfection facilities in 2010.	Still affected by polluted discharges from a small number of unsewered villages and expedient connections.
Sai Kung	Factory discharges are now controlled under the Water Pollution Control Ordinance. Most of the domestic and commercial premises in the urbanised town centers are connected to government sewers. The rural village houses are being connected to government sewers. Livestock farm discharges are now controlled under the Livestock Waste Control Scheme. The water quality of the rivers has improved from largely “Bad” in 1991 to largely “Excellent” in 2018. The <i>E. coli</i> level of rivers in Port Shelter WCZ has been reduced by over 70% compared with 1997. Sewerage works for villages are progressively being planned and implemented. Upgrading of the Sai Kung Sewage Treatment Works is under planning.	Still affected by polluted discharges from a small number of unsewered villages and expedient connections.

District	Improvement	Remaining Problem
New Territories		
Islands	Some of the domestic premises are connected to government sewers. Under the Livestock Waste Control Scheme, the small number of livestock farms has all been closed. Mui Wo River and Tung Chung River have stayed “Good” or “Excellent” throughout recent decades. The <i>E. coli</i> level of Mui Wo River has been reduced by over 90% compared with 1988. Additional village sewerage and sewage treatment facilities and upgrading of treatment facilities are being planned and provided to Southern Lantau Island, Tung Chung West, Mui Wo, Tai O, Lamma Island, Cheung Chau and Peng Chau.	Still affected by discharges from a small number of unsewered villages and some expedient connections.
Kowloon		
Yau Tsim Mong	The urban area of Kowloon contains few natural watercourses. Factory and restaurant discharges are now controlled under the Water Pollution Control Ordinance. Domestic and commercial premises are connected to government sewers. A large number of expedient connections have been rectified. The water quality of the Kai Tak River improved from mainly “Very Bad” in 1986 to all “Good” in 2018. The <i>E. coli</i> level of the region has been reduced by over 90% compared with 1999. The Upgrading of Central and East Kowloon Sewerage Phase 3 commenced in January 2019 to increase the sewerage capacity in Central Kowloon.	Mainly affected by rear lane washing activities, urban run-offs and expedient connections.
Sham Shui Po		
Kowloon City		
Wong Tai Sin		
Kwun Tong		
Hong Kong Island		
Central and Western	The urban area of Hong Kong Island contains few natural watercourses. Factory and restaurant discharges are now controlled under the Water Pollution Control Ordinance. Domestic and commercial premises are connected to government sewers. A large number of expedient connections have been rectified.	Mainly affected by rear lane washing activities, urban run-offs and expedient connections. Some streams are affected by discharges from squatters built near the streams.
Wan Chai		
Eastern		
Southern		